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SECOND ORDER RUNGE–KUTTA METHODS FOR ITÔ STOCHASTIC DIFFERENTIAL EQUATIONS*

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Abstract. A new class of stochastic Runge–Kutta methods for the weak approximation of the solution of Itô stochastic differential equation systems with a multidimensional Wiener process is introduced. As the main innovation, the number of stages of the methods does not depend on the dimension of the driving Wiener process, and the number of necessary random variables which have to be simulated is reduced considerably. Compared to well-known schemes, this reduces the computational effort significantly. Order conditions for the stochastic Runge–Kutta methods assuring weak convergence with order two are calculated by applying the colored rooted tree analysis due to the author. Further, some coefficients for explicit second order stochastic Runge–Kutta schemes are presented.

Key words. stochastic Runge–Kutta method, stochastic differential equation, multicolored rooted tree analysis, weak approximation, numerical method

AMS subject classifications. 65C30, 65L06, 60H35, 60H10

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1. Introduction.

The development of derivative-free approximation methods for solutions of stochastic differential equations (SDEs) has been the subject of recent research. For example, derivative-free Runge–Kutta-type schemes have been proposed for the strong approximation of solutions of SDEs in [1, 2, 7, 12, 14, 21]. On the other hand, for the weak approximation of solutions of SDEs, particular schemes have to be developed; see, e.g., [7, 8, 12, 15, 22]. Recently, second order stochastic Runge–Kutta (SRK) methods for the weak approximation have been studied by, e.g., Debrabant and Rößler [3], Kloeden and Platen [7], Komori et al. [9, 10], Rößler [18, 19, 16], and Tocino and Vigo-Aguiar [24]. However, to the author's knowledge, up to now all proposed second order SRK methods suffer from an inefficiency if they are applied to SDEs with a multidimensional Wiener process because the number of stages, and thus the number of evaluations of the diffusion function per step, depends at least linearly on the dimension m of the driving Wiener process. This drawback becomes significant especially for high-dimensional problems. The aim of the present paper is to overcome this drawback by introducing a new class of efficient SRK methods for Itô SDEs. Essentially, these new SRK methods possess two advantages: First, the number of random variables that have to be simulated is only $2m-1$ for each step, whereas the SRK methods proposed in, e.g., [3, 7, 24] need the simulation of $m(m+1)/2$ random variables. Second, in contrast to all present SRK methods for Itô SDEs the number of stages, and thus the number of evaluations of the drift and the diffusion functions per step, is constant for the new SRK methods, i.e., independent of the dimension m of the driving Wiener process. Recently, a class of efficient SRK methods with reduced complexity was proposed by the author for Stratonovich SDEs [20]. However, the generator of an Itô SDE differs significantly from that for a Stratonovich SDE [17]. Further, stochastic Itô–Taylor expansions based on Itô's formula turn out to be much

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more complex than in the case of stochastic Stratonovich–Taylor expansions where the rules of classical calculus can be applied [7]. Thus, the development of efficient SRK methods for Itô SDEs is a very challenging task because a specific design of the methods is needed which is in accordance with the Itô calculus. Since Itô calculus is widely used in many applications like, e.g., mathematical finance, there is particular interest in efficient SRK methods for Itô SDEs. Therefore, in the present paper we introduce efficient SRK methods designed specifically for Itô SDEs and calculate order conditions which turn out to differ significantly from the ones in the case of Stratonovich SDEs. Finally, the presented efficient SRK methods for Itô SDEs need only 3 stages, in contrast to the efficient SRK methods for Stratonovich SDEs in [20] where 4 stages are necessary. The paper is organized as follows: In sections 2–4 we briefly review the main concept of the rooted tree analysis for the weak approximation introduced in [18, 17]. Then, in section 5, a new class of SRK methods is introduced and the rooted tree analysis provided in sections 2–4 is applied to calculate order conditions. Further, some coefficients for explicit second order SRK schemes are presented. The paper closes with a numerical example in section 6.

Let $(\Omega, \mathcal{F}, P)$ be a probability space with a filtration $(\mathcal{F}_t)_{t\geq 0}$ fulfilling the usual conditions and let $\mathcal{I} = [t_0, T]$ for some $0 \leq t_0 < T < \infty$. We denote by $X = (X_t)_{t\in\mathcal{I}}$ the solution of the d-dimensional Itô SDE system

$$X_t = X_{t_0} + \int_{t_0}^{t} a(s, X_s)\,\mathrm{d}s + \sum_{j=1}^{m} \int_{t_0}^{t} b^j(s, X_s)\,\mathrm{d}W_s^j \tag{1.1}$$

for $d, m \geq 1$ with an m-dimensional Wiener process $(W_t)_{t\geq 0}$. Suppose $X_{t_0} = x_0$ is $\mathcal{F}_{t_0}$-measurable with $\mathrm{E}(\|X_{t_0}\|^{2l}) < \infty$ for some $l \in \mathbb{N}$. Throughout the paper we shall suppose that $a, b^j : \mathcal{I} \times \mathbb{R}^d \to \mathbb{R}^d$ for $j = 1, \ldots, m$ are at least Lipschitz continuous and satisfy a linear growth condition with respect to (w.r.t.) the state variable x and let a^i and $b^{i,j}$ denote the ith component for $1 \leq i \leq d$ of the d-dimensional vectors a and b^j. Then the existence and uniqueness theorem applies (see, e.g., [6]). In the following, let $C_P^l(\mathbb{R}^d, \mathbb{R})$ denote the space of all $g \in C^l(\mathbb{R}^d, \mathbb{R})$ with polynomial growth; i.e., there exists a constant $C > 0$ and $r \in \mathbb{N}$, such that $|\partial_x^i g(x)| \leq C(1 + \|x\|^{2r})$ for all $x \in \mathbb{R}^d$ and any partial derivative of order $i \leq l$ [7]. We say that g belongs to $C_P^{k,l}(\mathcal{I} \times \mathbb{R}^d, \mathbb{R})$ if $g \in C^{k,l}(\mathcal{I} \times \mathbb{R}^d, \mathbb{R})$ and $g(t, \cdot) \in C_P^l(\mathbb{R}^d, \mathbb{R})$ is fulfilled uniformly in $t \in \mathcal{I}$.

Let a discretization $\mathcal{I}_h = \{t_0, t_1, \ldots, t_N\}$ with $t_0 < t_1 < \ldots < t_N = T$ of the time interval $\mathcal{I} = [t_0, T]$ with step sizes $h_n = t_{n+1} - t_n$ for $n = 0, 1, \ldots, N-1$ be given. Further, let $h = \max_{0\leq n<N} h_n$ denote the maximum step size.

Definition 1.1. *A sequence of approximation processes $Y = (Y(t))_{t\in\mathcal{I}_h}$ converges weakly with order p to X at time T as $h \to 0$ if for each $f \in C_P^{2(p+1)}(\mathbb{R}^d, \mathbb{R})$ there exist a constant C_f and a finite $\delta_0 > 0$ such that*

$$|\,\mathrm{E}(f(X_T)) - \mathrm{E}(f(Y(T)))| \leq C_f\, h^p \tag{1.2}$$

for each $h \in \,]0, \delta_0[$.

2. Stochastic Runge–Kutta methods.

A very general class of SRK methods has been introduced in [18], which covers well-known schemes as well as many future schemes: Let $\mathcal{M}$ be a finite set of multi-indices with $\kappa = |\mathcal{M}|$ elements and let $\theta_\nu(h)$, $\nu \in \mathcal{M}$, be some random variables. Then a general class of s-stage SRK methods is given by $Y_0 = x_0$ and

$$Y_{n+1} = Y_n + \sum_{i=1}^{s} z_i^{(0,0)}\; a\left(t_n + c_i^{(0,0)} h_n, H_i^{(0,0)}\right) + \sum_{i=1}^{s}\sum_{k=1}^{m}\sum_{\nu\in\mathcal{M}} z_i^{(k,\nu)}\; b^k\left(t_n + c_i^{(k,\nu)} h_n, H_i^{(k,\nu)}\right) \tag{2.1}$$

for $n = 0, 1, \ldots, N-1$ with $Y_n = Y(t_n)$, $t_n \in \mathcal{I}_h$, and

$$H_i^{(k,\nu)} = Y_n + \sum_{j=1}^{s} Z_{ij}^{(k,\nu),(0,0)}\; a\left(t_n + c_j^{(0,0)} h_n, H_j^{(0,0)}\right) + \sum_{j=1}^{s}\sum_{r=1}^{m}\sum_{\mu\in\mathcal{M}} Z_{ij}^{(k,\nu),(r,\mu)}\; b^r\left(t_n + c_j^{(r,\mu)} h_n, H_j^{(r,\mu)}\right)$$

for $i = 1, \ldots, s$, $k = 0, 1, \ldots, m$, and $\nu \in \mathcal{M} \cup \{0\}$. Here, let

$$z_i^{(0,0)} = \alpha_i\, h_n, \qquad z_i^{(k,\nu)} = \sum_{\iota\in\mathcal{M}} {\gamma_i^{(\iota)}}^{(k,\nu)}\, \theta_\iota(h_n),$$

$$Z_{ij}^{(k,\nu),(0,0)} = A_{ij}^{(k,\nu),(0,0)}\, h_n, \qquad Z_{ij}^{(k,\nu),(r,\mu)} = \sum_{\iota\in\mathcal{M}} {B_{ij}^{(\iota)}}^{(k,\nu),(r,\mu)}\, \theta_\iota(h_n)$$

for $i, j = 1, \ldots, s$ and let $\alpha_i, {\gamma_i^{(\iota)}}^{(k,\nu)}, A_{ij}^{(k,\nu),(0,0)}, {B_{ij}^{(\iota)}}^{(k,\nu),(r,\mu)} \in \mathbb{R}$ be the coefficients of the SRK method. In the following, we always denote the corresponding vectors and matrices by, e.g., $z^{(k,\nu)} = (z_i^{(k,\nu)})_{1\le i\le s}$ and by $Z^{(k,\nu),(r,\mu)} = (Z_{ij}^{(k,\nu),(r,\mu)})_{1\le i,j\le s}$, where the subindices are omitted. Then the vector of weights can be defined by

$$c^{(k,\nu)} = A^{(k,\nu),(0,0)} e \tag{2.2}$$

with $e = (1, \ldots, 1)^T$. If $A_{ij}^{(k,\nu),(0,0)} = {B_{ij}^{(\iota)}}^{(k,\nu),(r,\mu)} = 0$ for $j \ge i$, then (2.1) is called an explicit SRK method; otherwise it is called implicit. We assume that the random variables $\theta_\nu(h_n)$ satisfy the moment condition

$$\mathrm{E}\left(\theta_{\nu_1}^{p_1}(h_n) \cdot \ldots \cdot \theta_{\nu_\kappa}^{p_\kappa}(h_n)\right) = \mathcal{O}\big(h_n^{(p_1+\ldots+p_\kappa)/2}\big) \tag{2.3}$$

for all $p_i \in \mathbb{N}_0$ and $\nu_i \in \mathcal{M}$, $1 \le i \le \kappa$. The moment condition ensures a contribution of each random variable having an order of magnitude $\mathcal{O}(\sqrt{h})$. This condition is in accordance with the order of magnitude of the increments of the Wiener process. Note that in the case of $b \equiv 0$, the SRK method reduces to the well-known deterministic Runge–Kutta method, so the introduced class of SRK methods turns out to be a generalization of deterministic Runge–Kutta methods.

3. Colored rooted tree analysis. Next, we give a short outline of the colored rooted tree analysis for the weak approximation following [18, 17]. Here, the trees need to be composed of three different types of nodes. Similar colored rooted trees based on only two different types of nodes have been applied in [2] for strong approximation methods. In section 5, the colored rooted tree analysis is applied for the calculation of order conditions for the new SRK methods, which are contained in the very general class (2.1). Without loss of generality, we restrict our considerations to autonomous d-dimensional Itô SDE systems with an m-dimensional Wiener process in this section. Let $TS(\Delta)$ denote the set of colored rooted trees with a root of type $\gamma = \otimes$ and which may additionally consist of some deterministic nodes of type $\tau = \bullet$ and stochastic nodes of type $\sigma_{j_k} = \bigcirc_{j_k}$ with a variable index $j_k \in \{1, \ldots, m\}$. The variable index j_k is associated with the j_kth component of the m-dimensional driving Wiener process of the considered SDE. If not stated otherwise, each stochastic node has its own variable index. So, if we have a tree with s stochastic nodes, then each stochastic node corresponds to exactly one of the indices $j_1, \ldots, j_s$.

In the following, let $l(\mathbf{t})$ be the number of nodes of $\mathbf{t} \in TS(\Delta)$. Then we denote by $d(\mathbf{t})$ the number of deterministic nodes and by $s(\mathbf{t})$ the number of stochastic nodes of $\mathbf{t} \in TS(\Delta)$, and we obtain $l(\mathbf{t}) = d(\mathbf{t}) + s(\mathbf{t}) + 1$. The order $\rho(\mathbf{t})$ of the tree $\mathbf{t} \in TS(\Delta)$ is defined as $\rho(\mathbf{t}) = d(\mathbf{t}) + \frac{1}{2} s(\mathbf{t})$ with $\rho(\gamma) = 0$. Following [18, 17], the trees are numbered by a subindex containing the order ρ of the tree followed by its identifier, which is the last number of the subindex. For example, $\rho(\mathbf{t}_{2,4}) = \rho(\mathbf{t}_{2,8}) = \rho(\mathbf{t}_{2,13}) = 2$.

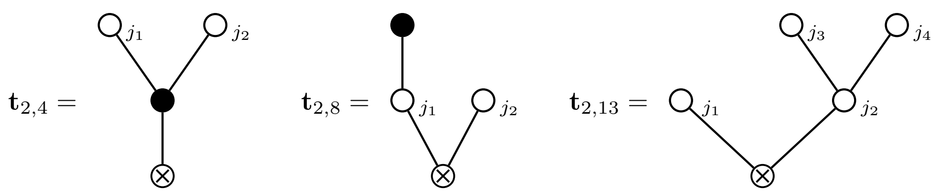

FIG. 3.1. *Some elements of* $TS(\Delta)$ *with* $j_1, j_2, j_3, j_4 \in \{1, \ldots, m\}$.

Every tree $\mathbf{t} \in TS(\Delta)$ can be written by a combination of three different brackets: If $\mathbf{t}_1, \ldots, \mathbf{t}_k$ are colored trees, then we denote by $(\mathbf{t}_1, \ldots, \mathbf{t}_k)$, $[\mathbf{t}_1, \ldots, \mathbf{t}_k]$, and $\{\mathbf{t}_1, \ldots, \mathbf{t}_k\}_j$ the tree in which $\mathbf{t}_1, \ldots, \mathbf{t}_k$ are each joined by a single branch to $\otimes$, $\bullet$, and $\bigcirc_j$, respectively. Here, the order of the subtrees $\mathbf{t}_1, \ldots, \mathbf{t}_k$ does not matter since any order leads to equivalent trees. Therefore, we obtain $\mathbf{t}_{2,4} = ([\bigcirc_{j_1}, \bigcirc_{j_2}]) = ([\sigma_{j_1}, \sigma_{j_2}])$, $\mathbf{t}_{2,8} = (\{\bullet\}_{j_1}, \bigcirc_{j_2}) = (\{\tau\}_{j_1}, \sigma_{j_2})$, and $\mathbf{t}_{2,13} = (\bigcirc_{j_1}, \{\bigcirc_{j_3}, \bigcirc_{j_4}\}_{j_2}) = (\sigma_{j_1}, \{\sigma_{j_3}, \sigma_{j_4}\}_{j_2})$ for the trees in Figure 3.1.

Now, let $LTS(\Delta)$ denote the set of monotonically labeled trees, i.e., where the nodes are monotonically numbered starting with number one at the root of the tree. Then $\alpha_\Delta(\mathbf{t})$ is the cardinality of $\mathbf{t}$, i.e., the number of possibilities of monotonically labeling the nodes of $\mathbf{t}$ with numbers $1, \ldots, l(\mathbf{t})$. For example, for $\mathbf{t}_{1,2} = (\sigma_{j_1}, \sigma_{j_2})$ there exists only one possible monotonically labeling $(\sigma_{j_1}^2, \sigma_{j_2}^3)^1$ and thus $\alpha_\Delta(\mathbf{t}_{1,2}) = 1$. In contrast to this, for $\mathbf{t}_{1.5,3} = (\tau, \sigma_{j_1})$ we obtain $\alpha_\Delta(\mathbf{t}_{1.5,3}) = 2$. Although $(\tau^2, \sigma_{j_1}^3)^1$ and $(\sigma_{j_1}^3, \tau^2)^1$ are equivalent trees, there exist two different labeled trees, $(\tau^2, \sigma_{j_1}^3)^1$ and $(\tau^3, \sigma_{j_1}^2)^1$. So one has to distinguish between the labels of deterministic and stochastic nodes (see [18] for details). Further, $\alpha_\Delta(\mathbf{t}_{2,4}) = 1$, $\alpha_\Delta(\mathbf{t}_{2,8}) = 3$, and $\alpha_\Delta(\mathbf{t}_{2,13}) = 4$.

We assign to each tree $\mathbf{t} \in TS(\Delta)$ an elementary differential which is defined recursively by $F(\gamma)(x) = f(x)$, $F(\tau)(x) = a(x)$, $F(\sigma_j)(x) = b^j(x)$, and

$$F(\mathbf{t})(x) = \begin{cases} f^{(k)}(x) \cdot (F(\mathbf{t}_1)(x), \ldots, F(\mathbf{t}_k)(x)) & \text{for } \mathbf{t} = (\mathbf{t}_1, \ldots, \mathbf{t}_k), \\ a^{(k)}(x) \cdot (F(\mathbf{t}_1)(x), \ldots, F(\mathbf{t}_k)(x)) & \text{for } \mathbf{t} = [\mathbf{t}_1, \ldots, \mathbf{t}_k], \\ b^{j^{(k)}}(x) \cdot (F(\mathbf{t}_1)(x), \ldots, F(\mathbf{t}_k)(x)) & \text{for } \mathbf{t} = \{\mathbf{t}_1, \ldots, \mathbf{t}_k\}_j. \end{cases} \tag{3.1}$$

Here $f^{(k)}$, $a^{(k)}$, and $b^{j^{(k)}}$ define a symmetric k-linear differential operator, and one can choose the sequence of subtrees $\mathbf{t}_1, \ldots, \mathbf{t}_k$ in an arbitrary order. For example, the

Ith component of $a^{(k)} \cdot (F(\mathbf{t}_1), \dots, F(\mathbf{t}_k))$ can be written as

$$(a^{(k)} \cdot (F(\mathbf{t}_1), \dots, F(\mathbf{t}_k)))^I = \sum_{J_1, \dots, J_k = 1}^{d} \frac{\partial^k a^I}{\partial x^{J_1} \dots \partial x^{J_k}} (F^{J_1}(\mathbf{t}_1), \dots, F^{J_k}(\mathbf{t}_k)),$$

where the components of vectors are denoted by superscript indices, which are chosen as capitals. For example, we obtain for $\mathbf{t}_{2,8}$ the elementary differential

$$F(\mathbf{t}_{2,8}) = f''(b^{j_1\prime}(a), b^{j_2}) = \sum_{J_1, J_2 = 1}^{d} \frac{\partial^2 f}{\partial x^{J_1} \partial x^{J_2}} \left(\sum_{K_1 = 1}^{d} \frac{\partial b^{J_1, j_1}}{\partial x^{K_1}} a^{K_1} \cdot b^{J_2, j_2} \right).$$

DEFINITION 3.1. *Let $TS(I)$ denote the set of trees $\boldsymbol{t} \in TS(\Delta)$ with a root of type γ which can be built by finite many steps of the following form:*

(a) *adding a deterministic node of type τ, or*

(b) *adding two stochastic nodes of type σ_{j_k}, both with the same new variable index j_k for some $k \in \mathbb{N}$, whereas neither of the two nodes is allowed to be directly connected by an edge with the other one.*

Let $LTS(I)$ denote the set of labeled trees $\boldsymbol{t} \in TS(I)$ with the nodes labeled in the same order as they are added. Then $\alpha_I(\boldsymbol{t})$ is the number of all possible different monotonically labels of $\boldsymbol{t} \in TS(I)$ with $\alpha_I(\boldsymbol{t}) = 0$ if $\boldsymbol{t} \notin TS(I)$.

For example, $(\{\sigma_{j_1}^3\}_{j_1}^2, \{\sigma_{j_2}^5\}_{j_2}^4)^1 \notin LTS(I)$, while $(\{\sigma_{j_2}^5\}_{j_1}^2, \{\sigma_{j_2}^4\}_{j_1}^3)^1 \in LTS(I)$. Here, we have to point out that in contrast to $TS(\Delta)$ each tree $\mathbf{t} \in TS(I)$ with $s(\mathbf{t})$ stochastic nodes has only $s(\mathbf{t})/2$ different variable indices $j_1, \dots, j_{s(\mathbf{t})/2} \in \{1, \dots, m\}$ because there are always pairs of two stochastic nodes with the same variable index because of Definition 3.1 (b). This is due to the structure of the generator for the solution process of an Itô SDE (1.1); see [17] for details. We obtain the following result due to [17, Theorem 3.2 and Proposition 5.1].

THEOREM 3.2. *For $p \in \mathbb{N}_0$, $f, a^i, b^{i,j} \in C_P^{2(p+1)}(\mathbb{R}^d, \mathbb{R})$, $i = 1, \dots, d$, $j = 1, \dots, m$, and for $t \in [t_0, T]$ with $h = t - t_0$ we have the truncated expansion*

$$\mathrm{E}^{t_0, x_0}(f(X_t)) = \sum_{\substack{\boldsymbol{t} \in TS(I) \\ \rho(\boldsymbol{t}) \le p}} \sum_{j_1, \dots, j_{s(\boldsymbol{t})/2} = 1}^{m} \frac{\alpha_I(\boldsymbol{t})\, F(\boldsymbol{t})(x_0)}{2^{s(\boldsymbol{t})/2} \rho(\boldsymbol{t})!} h^{\rho(\boldsymbol{t})} + \mathcal{O}(h^{p+1}). \tag{3.2}$$

Next, we give an expansion for the approximation process $(Y(t))_{t \in \mathcal{I}_h}$ defined by the SRK method (2.1). For $\mathbf{t} \in TS(\Delta)$ let the density $\gamma(\mathbf{t})$ be defined recursively by $\gamma(\mathbf{t}) = 1$ if $l(\mathbf{t}) = 1$ and

$$\gamma(\mathbf{t}) = \begin{cases} \displaystyle\prod_{i=1}^{\lambda} \gamma(\mathbf{t}_i) & \text{if } \mathbf{t} = (\mathbf{t}_1, \dots, \mathbf{t}_\lambda), \\ \displaystyle l(\mathbf{t}) \prod_{i=1}^{\lambda} \gamma(\mathbf{t}_i) & \text{if } \mathbf{t} = [\mathbf{t}_1, \dots, \mathbf{t}_\lambda] \text{ or } \mathbf{t} = \{\mathbf{t}_1, \dots, \mathbf{t}_\lambda\}_j. \end{cases}$$

Since the expansion for $(Y(t))_{t \in \mathcal{I}_h}$ contains the coefficients and the random variables of the SRK method, we define a coefficient function Φ_S which assigns to every tree

$\mathbf{t} \in TS(\Delta)$ an elementary weight:

$$\Phi_S(\mathbf{t}) = \begin{cases} \prod_{i=1}^{\lambda} \Phi_S(\mathbf{t}_i) & \text{if } \mathbf{t} = (\mathbf{t}_1, \ldots, \mathbf{t}_\lambda), \\ {z^{(0,0)}}^T \prod_{i=1}^{\lambda} \Psi^{(0,0)}(\mathbf{t}_i) & \text{if } \mathbf{t} = [\mathbf{t}_1, \ldots, \mathbf{t}_\lambda], \\ \sum_{\nu \in \mathcal{M}} {z^{(k,\nu)}}^T \prod_{i=1}^{\lambda} \Psi^{(k,\nu)}(\mathbf{t}_i) & \text{if } \mathbf{t} = \{\mathbf{t}_1, \ldots, \mathbf{t}_\lambda\}_k, \end{cases} \tag{3.3}$$

where $\Phi_S(\gamma) = 1$, $\Psi^{(k,\nu)}(\emptyset) = e$ with $\tau = [\emptyset]$, $\sigma_k = \{\emptyset\}_k$, and

$$\Psi^{(k,\nu)}(\mathbf{t}) = \begin{cases} Z^{(k,\nu),(0,0)} \prod_{i=1}^{\lambda} \Psi^{(0,0)}(\mathbf{t}_i) & \text{if } \mathbf{t} = [\mathbf{t}_1, \ldots, \mathbf{t}_\lambda], \\ \sum_{\mu \in \mathcal{M}} Z^{(k,\nu),(r,\mu)} \prod_{i=1}^{\lambda} \Psi^{(r,\mu)}(\mathbf{t}_i) & \text{if } \mathbf{t} = \{\mathbf{t}_1, \ldots, \mathbf{t}_\lambda\}_r. \end{cases} \tag{3.4}$$

Here $e = (1, \ldots, 1)^T$ and the product of vectors in (3.4) is defined by componentwise multiplication, i.e., $(a_1, \ldots, a_n) \cdot (b_1, \ldots, b_n) = (a_1 b_1, \ldots, a_n b_n)$. Note that $TS(I) \subset TS(\Delta)$. Further, each tree $\mathbf{t} \in TS(\Delta)$ has $s(\mathbf{t})$ different variable indices $j_1, \ldots, j_{s(t)}$, while a tree $\mathbf{u} \in TS(I)$ has only $s(\mathbf{u})/2$ different variable indices. Then Proposition 6.1 in [18] yields the following.

Proposition 3.3. *Let $(Y(t))_{t \in \mathcal{I}_h}$ be defined by the SRK method* (2.1). *Assume that the random variables fulfill $\theta_\iota(h) = \sqrt{h} \cdot \vartheta_\iota$ for $\iota \in \mathcal{M}$ with some bounded random variables ϑ_ι. Then for $p \in \mathbb{N}_0$, $f, a^i, b^{i,j} \in C_P^{2(p+1)}(\mathbb{R}^d, \mathbb{R})$ for $i = 1, \ldots, d$, $j = 1, \ldots, m$ and for $t \in [t_0, T]$ with $h = t - t_0$ we have the truncated expansion*

$$\mathrm{E}^{t_0,x_0}\left(f\left(Y(t)\right)\right) = \sum_{\substack{\boldsymbol{t} \in TS(\Delta) \\ \rho(\boldsymbol{t}) \leq p + \frac{1}{2}}} \sum_{j_1, \ldots, j_{s(t)}=1}^{m} \frac{\alpha_\Delta(\boldsymbol{t})\, \gamma(\boldsymbol{t})\, F(\boldsymbol{t})(x_0)\, \mathrm{E}\left(\Phi_S(\boldsymbol{t})\right)}{(l(\boldsymbol{t}) - 1)!} + \mathcal{O}(h^{p+1})\,. \tag{3.5}$$

4. Order conditions for stochastic Runge–Kutta methods. For the calculation of order conditions for the coefficients and random variables used by the SRK method (2.1), the truncated rooted tree expansions of the solution (3.2) and of the approximation (3.5) have to coincide. However, in (3.2) only trees in $TS(I)$ are applied, while in (3.5) trees in $TS(\Delta)$ are used. Since $TS(I) \subset TS(\Delta)$, we have to compare for each $\mathbf{t} \in TS(I)$ and a fixed choice of the indices $j_1, \ldots, j_{s(\mathbf{t})/2} \in \{1, \ldots, m\}$ (see Definition 3.1) the term $\frac{\alpha_I(\mathbf{t})\, F(\mathbf{t})(x_0)}{2^{s(\mathbf{t})/2}\, \rho(\mathbf{t})!}\, h^{\rho(\mathbf{t})}$ in (3.2) with the terms $\frac{\alpha_\Delta(\mathbf{u})\, \gamma(\mathbf{u})\, F(\mathbf{u})(x_0)\, \mathrm{E}(\Phi_S(\mathbf{u}))}{(l(\mathbf{u})-1)!}$ in (3.5) which belong to those trees $\mathbf{u} \in TS(\Delta)$ with indices $\hat{j}_1, \ldots, \hat{j}_{s(\mathbf{u})} \in \{1, \ldots, m\}$ which are equivalent to $\mathbf{t} \in TS(I)$.

For example, in Figure 4.1 the trees $\mathbf{t} = (\sigma_{j_1}, \sigma_{j_2}, \{\sigma_{j_2}\}_{j_1}) \in TS(I)$ and $\mathbf{u} = (\sigma_{\hat{j}_1}, \sigma_{\hat{j}_2}, \{\sigma_{\hat{j}_4}\}_{\hat{j}_3}) \in TS(\Delta)$ are equivalent except that $\mathbf{t}$ differs from $\mathbf{u}$ in the variable indices. However, for $j_1, j_2 \in \{1, \ldots, m\}$ fixed with $j_1 \neq j_2$, the trees $\mathbf{t}$ and $\mathbf{u}$ are equivalent if either $\hat{j}_1 = \hat{j}_3 = j_1$ and $\hat{j}_2 = \hat{j}_4 = j_2$ or if due to symmetry $\hat{j}_1 = \hat{j}_4 = j_2$ and $\hat{j}_2 = \hat{j}_3 = j_1$. Note that the trees $(\mathbf{t}_1, \ldots, \mathbf{t}_k)$ and $(\mathbf{t}_{\sigma(1)}, \ldots, \mathbf{t}_{\sigma(k)})$ are equivalent for any permutation σ of the set $\{1, \ldots, k\}$ (see section 3). Thus, if $j_1 \neq j_2$, then

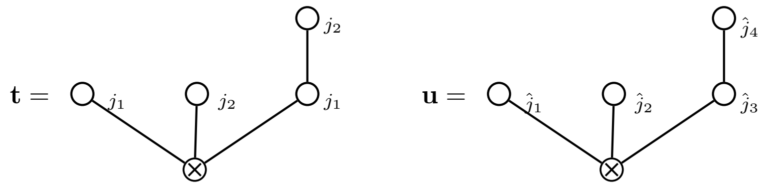

FIG. 4.1. *$\boldsymbol{t} \in TS(I)$ and $\boldsymbol{u} \in TS(\Delta)$ with $j_1, j_2, \hat{j}_1, \hat{j}_2, \hat{j}_3, \hat{j}_4 \in \{1, \ldots, m\}$.*

there exist two possible choices for the indices $\hat{j}_1, \hat{j}_2, \hat{j}_3$, and $\hat{j}_4$ such that $\mathbf{u} \in TS(\Delta)$ is equivalent to $\mathbf{t} \in TS(I)$. However, if $j_1 = j_2$, then the trees $\mathbf{t}$ and $\mathbf{u}$ are equivalent only if $\hat{j}_1 = \hat{j}_2 = \hat{j}_3 = \hat{j}_4 = j_1$. Here, we observe that for $\mathbf{t} \in TS(I)$ the number of equivalent trees in $TS(\Delta)$ does not depend on the particular choice of the indices $j_1, j_2 \in \{1, \ldots, m\}$ but only on the equivalence relation $j_1 = j_2$ or $j_1 \neq j_2$. Therefore, we consider the set of all binary equivalence relations over the sets $\{j_1, \ldots, j_{s(\mathbf{t})/2}\}$ and $\{\hat{j}_1, \ldots, \hat{j}_{s(\mathbf{u})}\}$, respectively. For example, an equivalence relation over $\{j_1, \ldots, j_{s(\mathbf{t})/2}\}$ declares either $j_k = j_l$ or $\neg(j_k = j_l)$ (i.e., $j_k \neq j_l$) for all $k, l \in \{1, \ldots, s(\mathbf{t})/2\}$. In the following, we denote for $\mathbf{t} \in TS(I)$ with a fixed equivalence relation over $j_1, \ldots, j_{s(\mathbf{t})/2}$ the number of all trees $\mathbf{u} \in TS(\Delta)$ with a fixed equivalence relation over $\{\hat{j}_1, \ldots, \hat{j}_{s(\mathbf{u})}\}$ which are equivalent to $\mathbf{t}$ by the combinatorial factor $\beta(\mathbf{t})$. As a result of this, we obtain in Figure 4.1 that $\beta(\mathbf{t}) = 2$ in the case of $j_1 \neq j_2$ and $\beta(\mathbf{t}) = 1$ if $j_1 = j_2$.

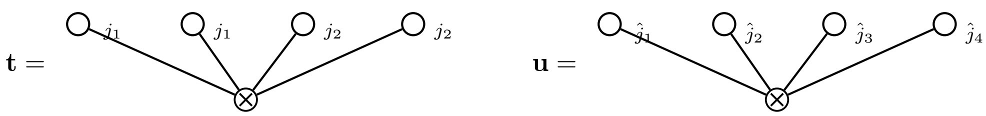

FIG. 4.2. *$\boldsymbol{t} \in TS(I)$ and $\boldsymbol{u} \in TS(\Delta)$ with $j_1, j_2, \hat{j}_1, \hat{j}_2, \hat{j}_3, \hat{j}_4 \in \{1, \ldots, m\}$.*

As a second example, consider in Figure 4.2 the trees $\mathbf{t} = (\sigma_{j_1}, \sigma_{j_1}, \sigma_{j_2}, \sigma_{j_2}) \in TS(I)$ and $\mathbf{u} = (\sigma_{\hat{j}_1}, \sigma_{\hat{j}_2}, \sigma_{\hat{j}_3}, \sigma_{\hat{j}_4}) \in TS(\Delta)$. In the case of $j_1 = j_2$, the tree $\mathbf{u}$ is equivalent to $\mathbf{t}$ if and only if the indices fulfill $\hat{j}_1 = \hat{j}_2 = \hat{j}_3 = \hat{j}_4 = j_1$. Thus there exists only one valid equivalence relation over $\{\hat{j}_1, \ldots, \hat{j}_{s(\mathbf{u})}\}$ for the tree $\mathbf{u}$, i.e., $\beta(\mathbf{t}) = 1$ if $j_1 = j_2$. However, in the case of $j_1 \neq j_2$, $\mathbf{u}$ is equivalent to $\mathbf{t}$ if either $\hat{j}_1 = \hat{j}_2 = j_1$ and $\hat{j}_3 = \hat{j}_4 = j_2$, if $\hat{j}_1 = \hat{j}_3 = j_1$ and $\hat{j}_2 = \hat{j}_4 = j_2$, or if $\hat{j}_1 = \hat{j}_4 = j_1$ and $\hat{j}_2 = \hat{j}_3 = j_2$. So, there exist 3 different valid equivalence relations over $\{\hat{j}_1, \hat{j}_2, \hat{j}_3, \hat{j}_4\}$ if $j_1 \neq j_2$. Note that the cases of $\hat{j}_1 = \hat{j}_2 = j_2$ and $\hat{j}_3 = \hat{j}_4 = j_1$, the case of $\hat{j}_1 = \hat{j}_3 = j_2$ and $\hat{j}_2 = \hat{j}_4 = j_1$, and also the case of $\hat{j}_1 = \hat{j}_4 = j_2$ and $\hat{j}_2 = \hat{j}_3 = j_1$ do not have to be taken into account additionally because they do not contribute any new equivalence relations over $\{\hat{j}_1, \ldots, \hat{j}_{s(\mathbf{u})}\}$. Only the number of different equivalence relations over $\{\hat{j}_1, \hat{j}_2, \hat{j}_3, \hat{j}_4\}$ which result in an equivalence of the trees $\mathbf{u}$ and $\mathbf{t}$ for suitable $\hat{j}_1, \hat{j}_2, \hat{j}_3, \hat{j}_4 \in \{1, \ldots, m\}$ has to be considered. Thus, $\beta(\mathbf{t}) = 3$ in the case of $j_1 \neq j_2$.

DEFINITION 4.1. *Let a tree $\boldsymbol{t} \in TS(I)$ with variable indices $j_1, \ldots, j_{s(\boldsymbol{t})/2}$ be given and let $\boldsymbol{u} \in TS(\Delta)$ with $s(\boldsymbol{u}) = s(\boldsymbol{t})$ and with variable indices $\hat{j}_1, \ldots, \hat{j}_{s(\boldsymbol{u})}$ denote a tree which is equivalent to $\boldsymbol{t}$ except that $\boldsymbol{u}$ differs from $\boldsymbol{t}$ in the variable indices. Then, for a fixed equivalence relation over $\{j_1, \ldots, j_{s(\boldsymbol{t})/2}\}$ of type $j_k = j_l$ or $j_k \neq j_l$, $1 \leq k < l \leq s(\boldsymbol{t})/2$, between the indices $j_1, \ldots, j_{s(\boldsymbol{t})/2}$ of $\boldsymbol{t}$, let $\beta(\boldsymbol{t})$ denote the number of all possible equivalence relations over $\{\hat{j}_1, \ldots, \hat{j}_{s(\boldsymbol{u})}\}$ of type $\hat{j}_k = \hat{j}_l$ or $\hat{j}_k \neq \hat{j}_l$, $1 \leq k < l \leq s(\boldsymbol{u})$, between the variable indices $\hat{j}_1, \ldots, \hat{j}_{s(\boldsymbol{u})}$ of $\boldsymbol{u}$ such that the tree $\boldsymbol{u}$ is* *equivalent to* $\boldsymbol{t}$ *also with respect to the variable indices. Further, for* $\boldsymbol{t} \in TS(\Delta)\backslash TS(I)$ *or in the case of* $s(\boldsymbol{t}) = 0$ *define* $\beta(\boldsymbol{t}) = 1$.

Note that in case of $m = 1$ we have $\beta(\mathbf{t}) = 1$ for all $\mathbf{t} \in TS(I)$. The following theorem yields conditions for the coefficients and the random variables of the SRK method (2.1) such that convergence with some order p in the weak sense is assured (see Theorem 6.4 in [18]).

THEOREM 4.2. *Let* $p \in \mathbb{N}$, $a^i, b^{i,j} \in C_P^{p+1,2p+2}(\mathcal{I} \times \mathbb{R}^d, \mathbb{R})$ *for* $i = 1, \ldots, d$, $j = 1, \ldots, m$. *Then the SRK method* (2.1) *with step size* h *is of weak order* p *if for all* $\boldsymbol{t} \in TS(\Delta)$ *with* $\rho(\boldsymbol{t}) \leq p + \frac{1}{2}$ *and all relations of type* $j_k = j_l$ *or* $j_k \neq j_l$, $1 \leq k < l \leq s(\boldsymbol{t})$, *between the indices* $j_1, \ldots, j_{s(\boldsymbol{t})} \in \{1, \ldots, m\}$ *of* $\boldsymbol{t}$ *the order conditions*

$$\mathrm{E}(\Phi_S(\boldsymbol{t})) = \frac{\alpha_I(\boldsymbol{t}) \cdot (l(\boldsymbol{t}) - 1)! \cdot h^{\rho(\boldsymbol{t})}}{\alpha_\Delta(\boldsymbol{t}) \cdot \beta(\boldsymbol{t}) \cdot \gamma(\boldsymbol{t}) \cdot 2^{s(\boldsymbol{t})/2} \cdot \rho(\boldsymbol{t})!} \tag{4.1}$$

are fulfilled, provided that (2.2) *and* (2.3) *apply and that the approximation* Y *has uniformly bounded moments w.r.t. the number* N *of steps.*

REMARK 4.3. *The approximation* Y *by the SRK method* (2.1) *has uniformly bounded moments if bounded random variables are used by the method, if* (2.3) *is fulfilled, and if* $\mathrm{E}({z^{(k,\nu)}}^T e) = 0$ *for* $1 \leq k \leq m$ *and* $\nu \in \mathcal{M}$ *(see* [18]*). Further, Theorem* 4.2 *provides uniform weak convergence with order* p *for all* $t \in \mathcal{I}_h$ *in the case of a nonrandom time discretization* $\mathcal{I}_h$ [18].

5. Order two stochastic Runge–Kutta methods for Itô SDEs.

5.1. Derivation of stochastic Runge–Kutta methods. In this section, we try to provide insight into the derivation of derivative-free weak approximation schemes. Further, we want to bring out the advantages of a newly introduced class of SRK methods compared to well-known SRK methods. Therefore, we start with the explicit weak order two derivative-free SRK scheme PL1WM due to Platen for autonomous Itô SDEs [3, 7, 24]. For $t_n \in \mathcal{I}_h$ with $Y_n = Y(t_n)$ the SRK method PL1WM for autonomous Itô SDEs is defined by $Y_0 = x_0$ and

$$\begin{aligned} Y_{n+1} = Y_n &+ \frac{1}{2}\left(a(Y_n) + a(H^{(0)})\right) h_n + \frac{1}{4}\sum_{k=1}^m \left(b^k(H_+^{(k)}) + b^k(H_-^{(k)}) + 2b^k(Y_n)\right) \hat{I}_{(k)} \\ &+ \frac{1}{4}\sum_{\substack{k,l=1\\k\neq l}}^m \left(b^k(\hat{H}_+^{(l)}) + b^k(\hat{H}_-^{(l)}) - 2b^k(Y_n)\right) \hat{I}_{(k)} \\ &+ \frac{1}{4}\sum_{k=1}^m \left(b^k(H_+^{(k)}) - b^k(H_-^{(k)})\right) \frac{\hat{I}_{(k,k)}}{\sqrt{h_n}} + \frac{1}{4}\sum_{\substack{k,l=1\\k\neq l}}^m \left(b^k(\hat{H}_+^{(l)}) - b^k(\hat{H}_-^{(l)})\right) \frac{\hat{I}_{(k,l)}}{\sqrt{h_n}} \end{aligned} \tag{5.1}$$

for $n = 0, 1, \ldots, N-1$ with stage values

$$H^{(0)} = Y_n + a(Y_n)\, h_n + \sum_{r=1}^m b^r(Y_n)\, \hat{I}_{(r)}\,, \qquad H_\pm^{(k)} = Y_n + a(Y_n)\, h_n \pm b^k(Y_n)\sqrt{h_n}\,,$$

$$\hat{H}_\pm^{(k)} = Y_n \pm b^k(Y_n)\sqrt{h_n}$$

for $k = 1, \dots, m$. The random variables are distributed with $\mathrm{P}(\hat{I}_{(k)} = \pm\sqrt{3\,h_n}) = \frac{1}{6}$ and $\mathrm{P}(\hat{I}_{(k)} = 0) = \frac{2}{3}$ for $k = 1, \dots, m$ and $\hat{J}_{(k,l)} = \hat{I}_{(k)}\hat{I}_{(l)} + V_{k,l}$ with $\mathrm{P}(V_{k,l} = \pm h_n) = \frac{1}{2}$ for $l = 1, \dots, k-1$, $V_{k,k} = -h_n$, and $V_{k,l} = -V_{l,k}$ for $l = k+1, \dots, m$ [7].

The scheme PL1WM (5.1) due to Platen has been derived from the order two weak Taylor scheme based on the truncated stochastic Itô–Taylor expansion (see, e.g., [7, 12, 17]) and is also contained in the class of SRK methods (2.1). In order to obtain the derivative-free SRK method (5.1), the derivatives of the drift and the diffusion functions a and b^k in the weak order two Taylor scheme are substituted by difference quotients such that the weak order two persists. For a detailed analysis of this class of SRK methods containing the scheme PL1WM, we refer to [3]. The computational complexity of the resulting scheme (5.1) can be determined by counting the number of evaluations of the functions a and b^k and by the number of random numbers needed for each step. As a result of this, the scheme PL1WM (5.1) due to Platen needs 2 evaluations of the drift a and $2m+1$ evaluations of each diffusion function b^k for $k = 1, \dots, m$. Further, the generation of $m(m+1)/2$ independent random numbers is necessary for each step of scheme (5.1). Thus, the SRK scheme (5.1) is not efficient if it is applied to SDEs with a high number m of driving Wiener processes due to the fact that the number of needed evaluations of each diffusion function grows linearly in m, which results in a very high computational complexity.

For the construction of a new class of efficient SRK methods with reduced complexity, the aim is to find a more efficient approximation of certain derivatives appearing in the stochastic Itô–Taylor expansion which are necessary for second order weak schemes. Following the idea of difference quotients as in the deterministic setting like in scheme (5.1), one obtains

$$\sum_{k,l=1}^{m} \frac{\partial b^k}{\partial x}(x_0)\, b^l(x_0)\, I_{(l,k)} \approx \sum_{k,l=1}^{m} \frac{b^k(x_0 + b^l(x_0)\sqrt{h}) - b^k(x_0 - b^l(x_0)\sqrt{h})}{2\sqrt{h}}\, I_{(l,k)} \tag{5.2}$$

as an approximation of the left term appearing in the Itô–Taylor expansion in the case of a $(d=1)$-dimensional solution process. In contrast to the deterministic case, one has to take care to assign the correct weight $I_{(l,k)}$ to the corresponding elementary differential $\frac{\partial b^k}{\partial x} b^l$. Considering the diffusion coefficient b^k for some arbitrarily fixed $k \in \{1, \dots, m\}$, we have to evaluate b^k at the values $x_0 + b^l(x_0)\sqrt{h}$ and $x_0 - b^l(x_0)\sqrt{h}$ for each $l = 1, \dots, m$ and also at x_0 in the case of $l = k$. Thus, $2m+1$ evaluations of each b^k, $k = 1, \dots, m$, are needed for this approach, which becomes very inefficient especially for large values of m. Therefore, the main idea for the design of a new efficient class of SRK methods is to use much more efficient approximations instead of deterministic difference quotients. Therefore, we propose, e.g.,

$$\begin{aligned} &\sum_{k,l=1}^{m} \frac{\partial b^k}{\partial x}(x_0)\, b^l(x_0)\, I_{(l,k)} \\ &\approx \sum_{k=1}^{m} \left(b^k\left(x_0 + \sum_{l=1}^{m} b^l(x_0)\, \frac{I_{(l,k)}}{\sqrt{h}} \right) - b^k\left(x_0 - \sum_{l=1}^{m} b^l(x_0)\, \frac{I_{(l,k)}}{\sqrt{h}} \right) \right) \frac{\sqrt{h}}{2} \end{aligned} \tag{5.3}$$

as a more efficient approximation which can be easily verified by standard Taylor expansion of the right-hand side considered as a multivariate function of $\sqrt{h}$ and $\frac{I_{(l,k)}}{\sqrt{h}}$ for $1 \le k, l \le m$. Again, one has to take care that the correct weight $I_{(l,k)}$ is assigned to the corresponding elementary differential $\frac{\partial b^k}{\partial x} b^l$. This is assured because calculating for some fixed $k,l \in \{1,\dots,m\}$ the mixed second derivative w.r.t. the variables $\frac{I_{(l,k)}}{\sqrt{h}}$ and $\sqrt{h}$ evaluated at $\frac{I_{(l,k)}}{\sqrt{h}} = 0$ for $1 \le k,l \le m$ and $\sqrt{h} = 0$, we obtain in the Taylor expansion $\frac{\partial b^k}{\partial x}(x_0) b^l(x_0)$ multiplied by the increments $\frac{I_{(l,k)}}{\sqrt{h}} \sqrt{h}$ and thus exactly the correct term as needed on the left-hand side of (5.3). The point is that the weight $\frac{I_{(l,k)}}{\sqrt{h}}$ appears only together with the corresponding composition of b^k and b^l, with b^k being the outer and b^l being the inner function. Considering the computational effort, for some arbitrarily fixed $k \in \{1,\dots,m\}$ the coefficient b^k has to be evaluated only 3 times, i.e., at the values $x_0 + \sum_{l=1}^m b^l(x_0) \frac{I_{(l,k)}}{\sqrt{h}}$, $x_0 - \sum_{l=1}^m b^l(x_0) \frac{I_{(l,k)}}{\sqrt{h}}$ and at x_0 if $l=k$. Thus, the number of needed evaluations no longer depends on the dimension m of the driving Wiener process. This is the reason why an approach of type (5.3) is much more efficient than (5.2), especially for high-dimensional problems. Additionally, we apply some different discrete random variables $\hat{I}_{(k,l)}$ for the weak approximation of the random variables $I_{(l,k)}$, which also need significantly reduced computational effort compared to the random variables $\hat{J}_{(k,l)}$ used in the scheme (5.1) due to Platen.

5.2. A new class of efficient stochastic Runge–Kutta methods. Now, we introduce a new class of second order SRK methods for the weak approximation of the solution of the Itô SDE (1.1) where the number of stages s is independent of the dimension m of the driving Wiener process. We define the d-dimensional approximation process Y with $Y_n = Y(t_n)$ for $t_n \in \mathcal{I}_h$ by the following SRK method with $Y_0 = x_0$ and

$$
\begin{aligned}
Y_{n+1} = Y_n &+ \sum_{i=1}^{s} \alpha_i \, a(t_n + c_i^{(0)} h_n, H_i^{(0)}) \, h_n \\
&+ \sum_{i=1}^{s} \sum_{k=1}^{m} \beta_i^{(1)} \, b^k(t_n + c_i^{(1)} h_n, H_i^{(k)}) \, \hat{I}_{(k)} \\
&+ \sum_{i=1}^{s} \sum_{k=1}^{m} \beta_i^{(2)} \, b^k(t_n + c_i^{(1)} h_n, H_i^{(k)}) \, \frac{\hat{I}_{(k,k)}}{\sqrt{h_n}} \\
&+ \sum_{i=1}^{s} \sum_{k=1}^{m} \beta_i^{(3)} \, b^k(t_n + c_i^{(2)} h_n, \hat{H}_i^{(k)}) \, \hat{I}_{(k)} \\
&+ \sum_{i=1}^{s} \sum_{k=1}^{m} \beta_i^{(4)} \, b^k(t_n + c_i^{(2)} h_n, \hat{H}_i^{(k)}) \, \sqrt{h_n}
\end{aligned}
\tag{5.4}
$$

for $n = 0, 1, \dots, N-1$ with stage values

$$
\begin{aligned}
H_i^{(0)} = Y_n &+ \sum_{j=1}^{s} A_{ij}^{(0)} \, a(t_n + c_j^{(0)} h_n, H_j^{(0)}) \, h_n \\
&+ \sum_{j=1}^{s} \sum_{l=1}^{m} B_{ij}^{(0)} \, b^l(t_n + c_j^{(1)} h_n, H_j^{(l)}) \, \hat{I}_{(l)},
\end{aligned}
$$

$$
\begin{aligned}
H_i^{(k)} &= Y_n + \sum_{j=1}^{s} A_{ij}^{(1)}\, a(t_n + c_j^{(0)} h_n, H_j^{(0)})\, h_n \\
&\quad + \sum_{j=1}^{s} B_{ij}^{(1)}\, b^k(t_n + c_j^{(1)} h_n, H_j^{(k)})\, \sqrt{h_n}, \\
\hat{H}_i^{(k)} &= Y_n + \sum_{j=1}^{s} A_{ij}^{(2)}\, a(t_n + c_j^{(0)} h_n, H_j^{(0)})\, h_n \\
&\quad + \sum_{j=1}^{s} \sum_{\substack{l=1\\ l\neq k}}^{m} B_{ij}^{(2)}\, b^l(t_n + c_j^{(1)} h_n, H_j^{(l)})\, \frac{\hat{I}_{(k,l)}}{\sqrt{h_n}}
\end{aligned}
$$

for $i = 1, \dots, s$ and $k = 1, \dots, m$. The random variables are defined by

$$
\hat{I}_{(k,l)} = \begin{cases} \frac{1}{2}(\hat{I}_{(k)}\hat{I}_{(l)} - \sqrt{h_n}\tilde{I}_{(k)}) & \text{if } k < l, \\ \frac{1}{2}(\hat{I}_{(k)}\hat{I}_{(l)} + \sqrt{h_n}\tilde{I}_{(l)}) & \text{if } l < k, \\ \frac{1}{2}(\hat{I}_{(k)}^2 - h_n) & \text{if } k = l \end{cases} \tag{5.5}
$$

for $1 \leq k, l \leq m$ with independent random variables $\hat{I}_{(k)}$, $1 \leq k \leq m$, such that

$$
\mathrm{E}(\hat{I}_{(k)}^q) = \begin{cases} 0 & \text{for } q \in \{1,3,5\}, \\ (q-1)h^{q/2} & \text{for } q \in \{2,4\}, \\ \mathcal{O}(h^{q/2}) & \text{for } q \geq 6 \end{cases} \tag{5.6}
$$

and random variables $\tilde{I}_{(k)}$, $1 \leq k \leq m-1$, possessing the moments

$$
\mathrm{E}(\tilde{I}_{(k)}^q) = \begin{cases} 0 & \text{for } q \in \{1,3\}, \\ h & \text{for } q = 2, \\ \mathcal{O}(h^{q/2}) & \text{for } q \geq 4. \end{cases} \tag{5.7}
$$

Thus, only $2m-1$ independent random variables are needed for each step. For example, we can choose $\hat{I}_{(k)}$ as three-point distributed random variables with $\mathrm{P}(\hat{I}_{(k)} = \pm\sqrt{3\,h_n}) = \frac{1}{6}$ and $\mathrm{P}(\hat{I}_{(k)} = 0) = \frac{2}{3}$. The random variables $\tilde{I}_{(k)}$ can be defined by a two-point distribution with $\mathrm{P}(\tilde{I}_{(k)} = \pm\sqrt{h_n}) = \frac{1}{2}$. The coefficients of the SRK method (5.4) can be represented by an extended Butcher array taking the form

$c^{(0)}$	$A^{(0)}$	$B^{(0)}$	
$c^{(1)}$	$A^{(1)}$	$B^{(1)}$	
$c^{(2)}$	$A^{(2)}$	$B^{(2)}$	
	α^T	${\beta^{(1)}}^T$	${\beta^{(2)}}^T$
		${\beta^{(3)}}^T$	${\beta^{(4)}}^T$

By calculating the order conditions due to Theorem 4.2, it turns out that there are some trees which restrict the structure of such SRK methods significantly. Therefore, we focus our investigations for a deeper insight on the exemplary trees

$$
\mathbf{t}_{2,12} = (\sigma_{j_1}, \sigma_{j_2}, \{\sigma_{j_4}\}_{j_3}), \qquad \mathbf{t}_{2,15} = (\{\sigma_{j_2}\}_{j_1}, \{\sigma_{j_4}\}_{j_3}), \tag{5.8}
$$

with $l(\mathbf{t}_{2,12}) = l(\mathbf{t}_{2,15}) = 5$, $\rho(\mathbf{t}_{2,12}) = \rho(\mathbf{t}_{2,15}) = 2$, and $s(\mathbf{t}_{2,12}) = s(\mathbf{t}_{2,15}) = 4$. We refer to the appendix for a detailed calculation of the complete order conditions. For the class of SRK methods (5.4) let

$$
\begin{aligned}
&z_i^{(0,0)} = \alpha_i\, h_n, && z_i^{(k,0)} = \beta_i^{(1)}\, \hat{I}_{(k)} + \beta_i^{(2)}\, \tfrac{\hat{I}_{(k,k)}}{\sqrt{h_n}}, && z_i^{(k,1)} = \beta_i^{(3)}\, \hat{I}_{(k)} + \beta_i^{(4)}\, \sqrt{h_n}, \\
&Z_{ij}^{(0,0)(0,0)} = A_{ij}^{(0)}\, h_n, && Z_{ij}^{(k,0)(0,0)} = A_{ij}^{(1)}\, h_n, && Z_{ij}^{(k,1)(0,0)} = A_{ij}^{(2)}\, h_n, \\
&Z_{ij}^{(0,0)(k,0)} = B_{ij}^{(0)} \hat{I}_{(k)}, && Z_{ij}^{(k,0)(l,0)} = B_{ij}^{(1)} \sqrt{h_n} \mathbf{1}_{\{k=l\}}, && Z_{ij}^{(k,1)(l,0)} = B_{ij}^{(2)} \tfrac{\hat{I}_{(k,l)}}{\sqrt{h_n}} \mathbf{1}_{\{k\neq l\}}
\end{aligned}
$$

for $1 \leq k, l \leq m$ and let $H_i^{(0,0)} = H_i^{(0)}$, $H_i^{(k,0)} = H_i^{(k)}$, and $H_i^{(k,1)} = \hat{H}_i^{(k)}$ for $1 \leq i, j \leq s$ (see also the appendix). Thus, the new class of SRK methods is contained in the very general class (2.1). Then the coefficient function (3.3) yields

$$
\begin{aligned}
\Phi_S(\mathbf{t}_{2,12}) &= (z^{(j_1,0)^T} e + z^{(j_1,1)^T} e)(z^{(j_2,0)^T} e + z^{(j_2,1)^T} e) \\
&\quad \times (z^{(j_3,0)^T} Z^{(j_3,0)(j_4,0)} e + z^{(j_3,1)^T} Z^{(j_3,1)(j_4,0)} e)\,, \\
\Phi_S(\mathbf{t}_{2,15}) &= (z^{(j_1,0)^T} Z^{(j_1,0)(j_2,0)} e + z^{(j_1,1)^T} Z^{(j_1,1)(j_2,0)} e) \\
&\quad \times (z^{(j_3,0)^T} Z^{(j_3,0)(j_4,0)} e + z^{(j_3,1)^T} Z^{(j_3,1)(j_4,0)} e)
\end{aligned} \tag{5.9}
$$

for $j_1, j_2, j_3, j_4 \in \{1, \ldots, m\}$. If we apply Theorem 4.2 to $\mathbf{t}_{2,12}$ and $\mathbf{t}_{2,15}$, then we have to consider the cases $j_k = j_l$ and $j_k \neq j_l$ for $1 \leq k < l \leq 4$. In the case of $j_1 = j_2 = j_3 = j_4$ we have $\alpha_I(\mathbf{t}_{2,12}) = 4$ and $\alpha_\Delta(\mathbf{t}_{2,12}) = 6$. Further, we obtain $\beta(\mathbf{t}_{2,12}) = 1$ as described in the example of section 4. As a result of this, the order condition (4.1) yields that $\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) = h_n^2$ has to be fulfilled. Applying (5.9) and taking into account the order conditions ${\beta^{(4)}}^T e = 0$ and ${\beta^{(2)}}^T e = 0$ due to the trees $\mathbf{t}_{0.5,1} = (\sigma_{j_1})$ and $\mathbf{t}_{1.5,4} = (\sigma_{j_1}, \sigma_{j_2}, \sigma_{j_3})$ (see the appendix for details) yields

$$
\begin{aligned}
\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) &= \mathrm{E}\Bigg((({\beta^{(1)}}^T e\, \hat{I}_{(j_1)} + {\beta^{(2)}}^T e\, \tfrac{\hat{I}_{(j_1,j_1)}}{\sqrt{h_n}}) + ({\beta^{(3)}}^T e\, \hat{I}_{(j_1)} + {\beta^{(4)}}^T e\, \sqrt{h_n}))^2 \\
&\quad \times ({\beta^{(1)}}^T B^{(1)} e\, \hat{I}_{(j_1)} \sqrt{h_n} + {\beta^{(2)}}^T B^{(1)} e\, \tfrac{\hat{I}_{(j_1,j_1)}}{\sqrt{h_n}} \sqrt{h_n}) \Bigg) \\
&= ({\beta^{(1)}}^T e + {\beta^{(3)}}^T e)^2 ({\beta^{(2)}}^T B^{(1)} e)\ \mathrm{E}(\hat{I}_{(j_1)}^2 \hat{I}_{(j_1,j_1)})\,.
\end{aligned}
$$

Due to $\mathrm{E}(\hat{I}_{(j_1)}^2 \hat{I}_{(j_1,j_1)}) = h_n^2$, the order condition is fulfilled if for the coefficients it holds that $({\beta^{(1)}}^T e + {\beta^{(3)}}^T e)^2 ({\beta^{(2)}}^T B^{(1)} e) = 1$. In the case of $j_1 = j_3 \neq j_2 = j_4$ we calculate with $\alpha_I(\mathbf{t}_{2,12}) = 4$, $\alpha_\Delta(\mathbf{t}_{2,12}) = 6$, and $\beta(\mathbf{t}_{2,12}) = 2$ from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) = \frac{1}{2} h_n^2$. Here, for the SRK method (5.4) we obtain

$$
\begin{aligned}
\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) &= \mathrm{E}\Bigg((({\beta^{(1)}}^T e\, \hat{I}_{(j_1)} + {\beta^{(2)}}^T e\, \tfrac{\hat{I}_{(j_1,j_1)}}{\sqrt{h_n}}) + ({\beta^{(3)}}^T e\, \hat{I}_{(j_1)} + {\beta^{(4)}}^T e\, \sqrt{h_n})) \\
&\quad \times (({\beta^{(1)}}^T e\, \hat{I}_{(j_2)} + {\beta^{(2)}}^T e\, \tfrac{\hat{I}_{(j_2,j_2)}}{\sqrt{h_n}}) + ({\beta^{(3)}}^T e\, \hat{I}_{(j_2)} + {\beta^{(4)}}^T e\, \sqrt{h_n})) \\
&\quad \times ({\beta^{(3)}}^T B^{(2)} e\, \hat{I}_{(j_1)} \tfrac{\hat{I}_{(j_1,j_2)}}{\sqrt{h_n}} + {\beta^{(4)}}^T B^{(2)} e\, \sqrt{h_n} \tfrac{\hat{I}_{(j_1,j_2)}}{\sqrt{h_n}}) \Bigg) \\
&= ({\beta^{(1)}}^T e + {\beta^{(3)}}^T e)^2 ({\beta^{(4)}}^T B^{(2)} e)\ \mathrm{E}(\hat{I}_{(j_1)} \hat{I}_{(j_2)} \hat{I}_{(j_1,j_2)})\,.
\end{aligned}
$$

Now, we can calculate that $\mathrm{E}(\hat{I}_{(j_1)}\hat{I}_{(j_2)}\hat{I}_{(j_1,j_2)}) = \frac{1}{2}h_n^2$. Thus, the order condition is fulfilled if $({\beta^{(1)}}^T e + {\beta^{(3)}}^T e)^2({\beta^{(4)}}^T B^{(2)} e) = 1$.

For $\mathbf{t}_{2,15}$, we calculate in the case of $j_1 = j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,15}) = 2$, $\alpha_\Delta(\mathbf{t}_{2,15}) = 3$, and $\beta(\mathbf{t}_{2,15}) = 1$ from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) = \frac{1}{2}h_n^2$. Again, applying (5.9) results in

$$\begin{aligned}\mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) &= \mathrm{E}(({\beta^{(1)}}^T B^{(1)} e\, \hat{I}_{(j_1)}\sqrt{h_n} + {\beta^{(2)}}^T B^{(1)} e\, \tfrac{\hat{I}_{(j_1,j_1)}}{\sqrt{h_n}}\sqrt{h_n})^2)\\ &= ({\beta^{(1)}}^T B^{(1)} e)^2\ \mathrm{E}(\hat{I}_{(j_1)}^2) h_n + ({\beta^{(2)}}^T B^{(1)} e)^2\ \mathrm{E}(\hat{I}_{(j_1,j_1)}^2)\,.\end{aligned}$$

Now, due to $\mathrm{E}(\hat{I}_{(j_1)}^2) = h_n$ and $\mathrm{E}(\hat{I}_{(j_1,j_1)}^2) = \frac{1}{2}h_n^2$ the order condition is $({\beta^{(1)}}^T B^{(1)} e)^2 + \frac{1}{2}({\beta^{(2)}}^T B^{(1)} e)^2 = \frac{1}{2}$. On the other hand, in the case of $j_1 = j_3 \neq j_2 = j_4$ with $\alpha_I(\mathbf{t}_{2,15}) = 2$, $\alpha_\Delta(\mathbf{t}_{2,15}) = 3$, and $\beta(\mathbf{t}_{2,15}) = 1$, we get from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) = \frac{1}{2}h_n^2$ has to be fulfilled. Now, we get from (5.9) that

$$\begin{aligned}\mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) &= \mathrm{E}(({\beta^{(3)}}^T B^{(2)} e\, \hat{I}_{(j_1)}\tfrac{\hat{I}_{(j_1,j_2)}}{\sqrt{h_n}} + {\beta^{(4)}}^T B^{(2)} e\, \sqrt{h_n}\tfrac{\hat{I}_{(j_1,j_2)}}{\sqrt{h_n}})^2)\\ &= ({\beta^{(3)}}^T B^{(2)} e)^2\ \mathrm{E}(\hat{I}_{(j_1)}^2 \hat{I}_{(j_1,j_2)}^2) h^{-1} + ({\beta^{(4)}}^T B^{(2)} e)^2\ \mathrm{E}(\hat{I}_{(j_1,j_2)}^2)\,.\end{aligned}$$

With $\mathrm{E}(\hat{I}_{(j_1)}^2 \hat{I}_{(j_1,j_2)}^2) = h_n^3$ and $\mathrm{E}(\hat{I}_{(j_1,j_2)}^2) = \frac{1}{2}h_n^2$, we finally get the order condition $({\beta^{(3)}}^T B^{(2)} e)^2 + \frac{1}{2}({\beta^{(4)}}^T B^{(2)} e)^2 = \frac{1}{2}$.

For all remaining cases of type $j_k = j_l$ or $j_k \neq j_l$ for $1 \leq k < l \leq 4$, we have $\alpha_I(\mathbf{t}_{2,12}) = \alpha_I(\mathbf{t}_{2,15}) = 0$ and $\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) = \mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) = 0$. Therefore, (4.1) is fulfilled in these cases without any additional restrictions for the coefficients.

The resulting order conditions from trees $\mathbf{t}_{2,12}$ and $\mathbf{t}_{2,15}$ involve terms of types ${\beta^{(1)}}^T B^{(1)} e$, ${\beta^{(2)}}^T B^{(1)} e$, ${\beta^{(3)}}^T B^{(2)} e$, and ${\beta^{(4)}}^T B^{(2)} e$. These order conditions are of special importance since they are directly affected by the modifications in the stage values of the new SRK method (5.4) compared to the SRK scheme (5.1). So, one has to ensure that the resulting order conditions admit a solution for the coefficients, which is the case for the new class of SRK methods (5.4) with reduced computational complexity. Applying the rooted tree analysis and Theorem 4.2 to all rooted trees up to order 2.5, we can calculate the following complete order two conditions for the SRK method (5.4) with the product of vectors defined by componentwise multiplication.

THEOREM 5.1. *Let $a^i, b^{i,j} \in C_P^{2,4}(\mathcal{I} \times \mathbb{R}^d, \mathbb{R})$ for $1 \leq i \leq d$, $1 \leq j \leq m$. If the coefficients of the stochastic Runge–Kutta method* (5.4) *fulfill the following equations, then the method attains order* 1 *for the weak approximation of the solution of the Itô SDE* (1.1)*:*

1. $\alpha^T e = 1$	2. ${\beta^{(4)}}^T e = 0$	3. ${\beta^{(3)}}^T e = 0$
4. $({\beta^{(1)}}^T e)^2 = 1$	5. ${\beta^{(2)}}^T e = 0$	6. ${\beta^{(1)}}^T B^{(1)} e = 0$
7. ${\beta^{(4)}}^T A^{(2)} e = 0$	8. ${\beta^{(3)}}^T B^{(2)} e = 0$	9. ${\beta^{(4)}}^T (B^{(2)} e)^2 = 0$

Further, if $a^i, b^{i,j} \in C_P^{3,6}(\mathcal{I} \times \mathbb{R}^d, \mathbb{R})$ for $1 \leq i \leq d$, $1 \leq j \leq m$ and if in addition the following equations are fulfilled and if $c^{(i)} = A^{(i)} e$ for $i = 0, 1, 2$, then the stochastic

Runge–Kutta method (5.4) *attains order* 2 *for the weak approximation of the solution of the Itô SDE* (1.1)*:*

10. $\alpha^T A^{(0)} e = \frac{1}{2}$
11. $\alpha^T (B^{(0)} e)^2 = \frac{1}{2}$
12. $(\beta^{(1)^T} e)(\alpha^T B^{(0)} e) = \frac{1}{2}$
13. $(\beta^{(1)^T} e)(\beta^{(1)^T} A^{(1)} e) = \frac{1}{2}$
14. $\beta^{(3)^T} A^{(2)} e = 0$
15. $\beta^{(2)^T} B^{(1)} e = 1$
16. $\beta^{(4)^T} B^{(2)} e = 1$
17. $(\beta^{(1)^T} e)(\beta^{(1)^T} (B^{(1)} e)^2) = \frac{1}{2}$
18. $(\beta^{(1)^T} e)(\beta^{(3)^T} (B^{(2)} e)^2) = \frac{1}{2}$
19. $\beta^{(1)^T} (B^{(1)} (B^{(1)} e)) = 0$
20. $\beta^{(3)^T} (B^{(2)} (B^{(1)} e)) = 0$
21. $\beta^{(3)^T} (B^{(2)} (B^{(1)} (B^{(1)} e))) = 0$
22. $\beta^{(1)^T} (A^{(1)} (B^{(0)} e)) = 0$
23. $\beta^{(3)^T} (A^{(2)} (B^{(0)} e)) = 0$
24. $\beta^{(4)^T} (A^{(2)} e)^2 = 0$
25. $\beta^{(4)^T} (A^{(2)} (A^{(0)} e)) = 0$
26. $\alpha^T (B^{(0)} (B^{(1)} e)) = 0$
27. $\beta^{(2)^T} A^{(1)} e = 0$
28. $\beta^{(1)^T} ((A^{(1)} e)(B^{(1)} e)) = 0$
29. $\beta^{(3)^T} ((A^{(2)} e)(B^{(2)} e)) = 0$
30. $\beta^{(4)^T} (A^{(2)} (B^{(0)} e)) = 0$
31. $\beta^{(2)^T} (A^{(1)} (B^{(0)} e)) = 0$
32. $\beta^{(4)^T} ((B^{(2)} e)^2 (A^{(2)} e)) = 0$
33. $\beta^{(4)^T} (A^{(2)} (B^{(0)} e)^2) = 0$
34. $\beta^{(2)^T} (A^{(1)} (B^{(0)} e)^2) = 0$
35. $\beta^{(1)^T} (B^{(1)} (A^{(1)} e)) = 0$
36. $\beta^{(3)^T} (B^{(2)} (A^{(1)} e)) = 0$
37. $\beta^{(2)^T} (B^{(1)} e)^2 = 0$
38. $\beta^{(4)^T} (B^{(2)} (B^{(1)} e)) = 0$
39. $\beta^{(2)^T} (B^{(1)} (B^{(1)} e)) = 0$
40. $\beta^{(1)^T} (B^{(1)} e)^3 = 0$
41. $\beta^{(3)^T} (B^{(2)} e)^3 = 0$
42. $\beta^{(1)^T} (B^{(1)} (B^{(1)} e)^2) = 0$
43. $\beta^{(3)^T} (B^{(2)} (B^{(1)} e)^2) = 0$
44. $\beta^{(4)^T} (B^{(2)} e)^4 = 0$
45. $\beta^{(4)^T} (B^{(2)} (B^{(1)} e))^2 = 0$
46. $\beta^{(4)^T} ((B^{(2)} e)(B^{(2)} (B^{(1)} e))) = 0$
47. $\alpha^T ((B^{(0)} e)(B^{(0)} (B^{(1)} e))) = 0$
48. $\beta^{(1)^T} ((A^{(1)} (B^{(0)} e))(B^{(1)} e)) = 0$
49. $\beta^{(3)^T} ((A^{(2)} (B^{(0)} e))(B^{(2)} e)) = 0$
50. $\beta^{(1)^T} (A^{(1)} (B^{(0)} (B^{(1)} e))) = 0$
51. $\beta^{(3)^T} (A^{(2)} (B^{(0)} (B^{(1)} e))) = 0$
52. $\beta^{(4)^T} ((B^{(2)} (A^{(1)} e))(B^{(2)} e)) = 0$
53. $\beta^{(1)^T} (B^{(1)} (A^{(1)} (B^{(0)} e))) = 0$
54. $\beta^{(3)^T} (B^{(2)} (A^{(1)} (B^{(0)} e))) = 0$
55. $\beta^{(1)^T} ((B^{(1)} e)(B^{(1)} (B^{(1)} e))) = 0$
56. $\beta^{(3)^T} ((B^{(2)} e)(B^{(2)} (B^{(1)} e))) = 0$
57. $\beta^{(1)^T} (B^{(1)} (B^{(1)} (B^{(1)} e))) = 0$
58. $\beta^{(4)^T} ((B^{(2)} e)(B^{(2)} (B^{(1)} (B^{(1)} e)))) = 0$
59. $\beta^{(4)^T} ((B^{(2)} e)(B^{(2)} (B^{(1)} e)^2)) = 0$

The proof of Theorem 5.1 can be found in the appendix.

REMARK 5.2. *Due to Theorem* 5.1 *we have to solve* 59 *equations for* $m > 1$. *However, in the case of* $m = 1$ *and if we choose* $A_{ij}^{(2)} = 0$ *for* $1 \leq i, j \leq s$, *then the* 59 *conditions reduce to* 28 *conditions (see also* [19]*). For an explicit SRK method of type* (5.4), $s \geq 3$ *is needed due to conditions* 4, 6, *and* 17, *which cannot be fulfilled for* $s < 3$. *Note that in the case of commutative noise, significantly simplified SRK methods have been developed which need only* m *independent random variables for each step; see* [16] *for details.*

Considering the order conditions 1–9 of Theorem 5.1, we can easily calculate SRK schemes converging with order one in the weak sense. For example, the well-known Euler–Maruyama scheme EM (see, e.g., [7]) belongs to the introduced class of SRK methods having order 1 with $s = 1$ stage and with coefficients $\alpha_1 = \beta_1^{(1)} = 1$, $\beta_1^{(2)} = \beta_1^{(3)} = \beta_1^{(4)} = 0$, $A_{11}^{(0)} = A_{11}^{(1)} = 0$, and $B_{11}^{(0)} = B_{11}^{(1)} = 0$. Further, if we calculate order two SRK schemes with $s \geq 3$ stages, then there are some degrees of freedom in choosing the coefficients. We refer to [4] for a detailed analysis of the solution space of the order conditions in Theorem 5.1 and for some coefficients which minimize the error constants of the SRK method (5.4). Especially, it is possible to calculate an SRK scheme converging with some higher order if it is applied to a deterministic ordinary differential equation. For example, if the weights α_i and the coefficients $A_{ij}^{(0)}$ are chosen such that the conditions of Theorem 5.1 and additionally the conditions $\alpha^T(A^{(0)}(A^{(0)}e)) = \frac{1}{6}$ and $\alpha^T(A^{(0)}e)^2 = \frac{1}{3}$ are fulfilled (see, e.g., [5]), then the SRK scheme is of order three in the case of $b^j \equiv 0$ for $1 \leq j \leq m$ in SDE (1.1). Therefore, let (p_D, p_S) with $p_D \geq p_S$ denote the order of convergence of the SRK scheme if it is applied to a deterministic or stochastic differential equation, respectively. Thus, the scheme converges at least with order $p = p_S$ in the weak sense, and we suppose better convergence for schemes with $p_D > p_S$, particularly for SDEs with small noise. The SRK schemes RI1, RI3, and RI5 presented in Tables 5.1 and 5.2, respectively, are of order $p_S = 2$ and $p_D = 3$, while the SRK scheme RI6 in Table 5.2 is of order $p_D = p_S = 2$. Note that the schemes RI1, RI3, RI5, and RI6 coincide with the SRK schemes RI1W1, RI3W1, RI5W1, and PL1W1, respectively, proposed in [19] in the case that they are applied to SDEs with scalar noise, i.e., in the case of $d \geq 1$ and $m = 1$, due to $A_{ij}^{(2)} = 0$ for $1 \leq i, j \leq s$.

Considering the computational complexity of the new SRK schemes RI1, RI3, RI5, and RI6, we take again the number of evaluations of the drift and diffusion

TABLE 5.1
*SRK scheme RI*1 *and RI*3 *of order* $p_D = 3$ *and* $p_S = 2$.

$$
\begin{array}{c|ccc|ccc|ccc}
0 & & & & & & & & & \\
\frac{2}{3} & \frac{2}{3} & & & 1 & & & & & \\
\frac{2}{3} & -\frac{1}{3} & 1 & & 0 & 0 & & & & \\
\hline
0 & & & & & & & & & \\
1 & 1 & & & 1 & & & & & \\
1 & 1 & 0 & & -1 & 0 & & & & \\
\hline
0 & & & & & & & & & \\
0 & 0 & & & 1 & & & & & \\
0 & 0 & 0 & & -1 & 0 & & & & \\
\hline
 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} & -\frac{1}{2} \\
\cline{2-10}
 & & & & -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} & -\frac{1}{2}
\end{array}
\qquad
\begin{array}{c|ccc|ccc|ccc}
0 & & & & & & & & & \\
1 & 1 & & & \frac{3-2\sqrt{6}}{5} & & & & & \\
\frac{1}{2} & \frac{1}{4} & \frac{1}{4} & & \frac{6+\sqrt{6}}{10} & 0 & & & & \\
\hline
0 & & & & & & & & & \\
1 & 1 & & & 1 & & & & & \\
1 & 1 & 0 & & -1 & 0 & & & & \\
\hline
0 & & & & & & & & & \\
0 & 0 & & & 1 & & & & & \\
0 & 0 & 0 & & -1 & 0 & & & & \\
\hline
 & \frac{1}{6} & \frac{1}{6} & \frac{2}{3} & \frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} & -\frac{1}{2} \\
\cline{2-10}
 & & & & -\frac{1}{2} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{2} & -\frac{1}{2}
\end{array}
$$

Table 5.2
SRK scheme RI5 of order $p_D = 3$ and $p_S = 2$ and RI6 of order $p_D = p_S = 2$.

0									
1	1			$\frac{1}{3}$					
$\frac{5}{12}$	$\frac{25}{144}$	$\frac{35}{144}$		$-\frac{5}{6}$	0				
0									
$\frac{1}{4}$	$\frac{1}{4}$			$\frac{1}{2}$					
$\frac{1}{4}$	$\frac{1}{4}$	0		$-\frac{1}{2}$	0				
0									
0	0			1					
0	0	0		-1	0				
	$\frac{1}{10}$	$\frac{3}{14}$	$\frac{24}{35}$	1	-1	-1	0	1	-1
				$\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{4}$	0	$\frac{1}{2}$	$-\frac{1}{2}$

0									
1	1			1					
0	0	0		0	0				
0									
1	1			1					
1	1	0		-1	0				
0									
0	0			1					
0	0	0		-1	0				
	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	0	$\frac{1}{2}$	$-\frac{1}{2}$
				$-\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	0	$\frac{1}{2}$	$-\frac{1}{2}$

functions and the number of random numbers needed in each step as a measure for the complexity of the schemes. Then we obtain the following result: the SRK schemes RI1, RI3, and RI5 need 3 evaluations of the drift a, while the SRK scheme RI6 needs 2 evaluations of a. Furthermore, we have to point out that only 5 evaluations of each diffusion function b^k for $k = 1, \ldots, m$ are needed by all of the new SRK schemes RI1, RI3, RI5, and RI6. This is due to the fact that the number of stages $s = 3$ no longer depends on the dimension m of the driving Wiener process and because of $H_1^{(k)} = \hat{H}_1^{(k)}$, which saves one evaluation of each b^k in the case of explicit SRK schemes. As a further feature, only $2m - 1$ independent random numbers have to be generated for the new SRK schemes in each step. This is a significant reduction of complexity for the new SRK method (5.4) compared to the well-known SRK methods of type (5.1).

6. Numerical example. In order to verify the theoretical results, we compare the SRK schemes RI5 and RI6 to the order one Euler–Maruyama scheme EM, to the second order SRK scheme PL1WM (5.1) due to Platen [7], which is also considered in [24], and to the extrapolated Euler–Maruyama scheme ExEu due to Talay and Tubaro [23] attaining order two. The extrapolated Euler–Maruyama scheme ExEu scheme is given by $2\,\mathrm{E}(f(Z_T^{h/2})) - \mathrm{E}(f(Z_T^h))$ based on the EM approximations $Z_T^{h/2}$ and Z_T^h calculated with step sizes h and $h/2$. In the following, we approximate $\mathrm{E}(f(X_T))$ for $f(x^1, x^2) = x^1$, $f(x^1, x^2) = x^1\, x^2$, and $f(x^1, x^2) = (x^1)^2$ by Monte Carlo simulation. Therefore, we estimate $\mathrm{E}(f(Y_T))$ by the sample average of M independently simulated realizations of the approximations $f(Y_{T,k})$, $k = 1, \ldots, M$, with $Y_{T,k}$ calculated by the scheme under consideration. Then the error is denoted by $\hat{\mu} = E(f(X_T)) - \frac{1}{M} \sum_{k=1}^{M} f(Y_{T,k})$. The empirical variance $\hat{\sigma}_\mu^2$ of the error $\hat{\mu}$ is calculated following [7] based on M_1 batches with M_2 trajectories in each, i.e., $M = M_1 \cdot M_2$. Since we want to investigate the systematic error of the considered schemes, we have to minimize the statistical error by choosing M very large [7], i.e., much larger than usually necessary for an approximation in practice. The obtained errors at time $T = 1.0$ are plotted versus the corresponding step sizes or the corresponding computational effort with double logarithmic scale in order to analyze the empirical order of convergence and the performance of the schemes, respectively. In some figures, dotted reference lines with slopes 1.0 and 2.0 are plotted for comparison.

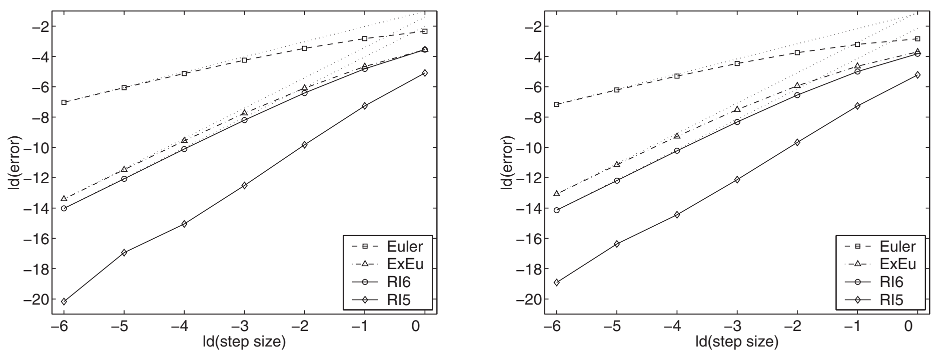

FIG. 6.1. *Step size vs. error for the approximation of* $\mathrm{E}(X_T^1)$ *(left) and of* $\mathrm{E}((X_T^1)^2)$ *(right) for SDE* (6.1).

TABLE 6.1
Results for the approximation of $\mathrm{E}(X_t^1)$ *for SDE* (6.1).

	RI5		RI6		EM		ExEu		PL1WM	
h	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$
2^{-0}	2.94e-2	1.11e-10	8.57e-2	1.21e-10	1.98e-1	1.95e-11	8.57e-2	1.19e-10	8.57e-2	1.21e-10
2^{-1}	6.52e-3	1.85e-10	3.56e-2	1.85e-10	1.42e-1	4.40e-11	3.95e-2	1.72e-10	3.56e-2	1.85e-10
2^{-2}	1.10e-3	3.48e-10	1.18e-2	3.43e-10	9.07e-2	1.33e-10	1.48e-2	3.27e-10	1.18e-2	3.43e-10
2^{-3}	1.71e-4	1.57e-10	3.40e-3	1.71e-10	5.27e-2	9.67e-11	4.67e-3	1.64e-10	3.40e-3	1.71e-10
2^{-4}	2.95e-5	4.16e-10	9.08e-4	4.41e-10	2.87e-2	3.25e-10	1.33e-3	4.35e-10	9.08e-4	4.41e-10
2^{-5}	7.93e-6	7.97e-10	2.32e-4	8.51e-10	1.50e-2	7.25e-10	3.52e-4	8.43e-10	2.32e-4	8.51e-10
2^{-6}	8.47e-7	6.23e-10	6.01e-5	6.18e-10	7.69e-3	5.71e-10	9.23e-5	6.16e-10	6.01e-5	6.18e-10

TABLE 6.2
Results for the approximation of $\mathrm{E}((X_t^1)^2)$ *for SDE* (6.1).

	RI5		RI6		EM		ExEu		PL1WM	
h	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$
2^{-0}	2.70e-2	7.91e-11	7.08e-2	6.32e-11	1.40e-1	4.84e-12	7.73e-2	5.13e-11	7.08e-2	6.32e-11
2^{-1}	6.51e-3	1.44e-10	3.14e-2	1.31e-10	1.09e-1	1.69e-11	4.01e-2	1.05e-10	3.14e-2	1.31e-10
2^{-2}	1.23e-3	2.85e-10	1.07e-2	2.72e-10	7.44e-2	6.97e-11	1.64e-2	2.40e-10	1.07e-2	2.72e-10
2^{-3}	2.24e-4	1.26e-10	3.12e-3	1.48e-10	4.54e-2	6.50e-11	5.50e-3	1.38e-10	3.12e-3	1.48e-10
2^{-4}	4.49e-5	3.34e-10	8.37e-4	3.74e-10	2.54e-2	2.40e-10	1.62e-3	3.65e-10	8.37e-4	3.74e-10
2^{-5}	1.18e-5	6.40e-10	2.14e-4	7.26e-10	1.35e-2	5.76e-10	4.39e-4	7.17e-10	2.14e-4	7.26e-10
2^{-6}	2.03e-6	5.14e-10	5.55e-5	5.06e-10	6.99e-3	4.51e-10	1.16e-4	5.04e-10	5.55e-5	5.06e-10

First, we consider for $d = m = 2$ the linear SDE system with commutative noise:

$$\mathrm{d}\begin{pmatrix} X_t^1 \\ X_t^2 \end{pmatrix} = \begin{pmatrix} \frac{3}{2}X_t^1 \\ \frac{3}{2}X_t^2 \end{pmatrix} \mathrm{d}t + \begin{pmatrix} \frac{1}{10}X_t^1 \\ 0 \end{pmatrix} \mathrm{d}W_t^1 + \begin{pmatrix} 0 \\ \frac{1}{10}X_t^2 \end{pmatrix} \mathrm{d}W_t^2, \tag{6.1}$$

with initial value $X_0 = (\frac{1}{10}, \frac{1}{10})^T$. The moments of the solution are given as $\mathrm{E}(X_t^i) = \frac{1}{10}\exp(\frac{3}{2}t)$ and $E((X_t^i)^2) = \frac{1}{100}\exp(\frac{301}{100}t)$ for $i = 1, 2$. Here, we choose $M_1 = 20$ batches with $M_2 = 5 \times 10^6$ trajectories in each. The errors $|\hat{\mu}|$ and empirical variances $\hat{\sigma}_\mu^2$ with corresponding step sizes are presented in Figure 6.1 and Tables 6.1–6.2.

The second test equation is a nonlinear SDE system for $d = m = 2$ with noncom-

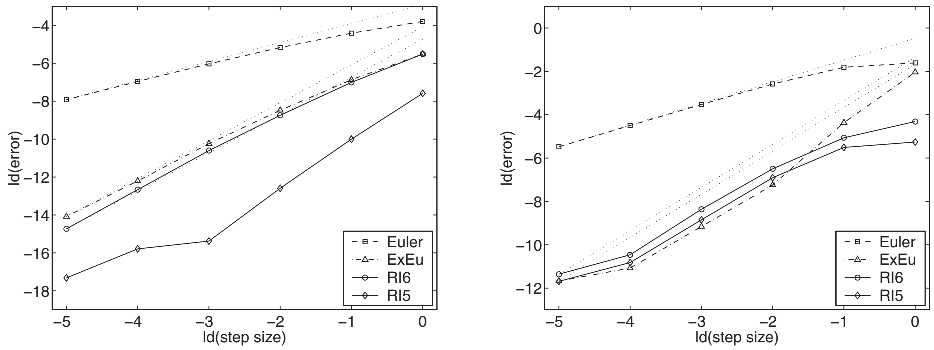

FIG. 6.2. *Step size vs. error for the approximation of* $\mathrm{E}(X_T^1)$ *(left) and of* $\mathrm{E}(X_T^1 X_T^2)$ *(right) for SDE* (6.2).

TABLE 6.3
Results for the approximation of $\mathrm{E}(X_t^1)$ *for SDE* (6.2).

	RI5		RI6		EM		ExEu		PL1WM	
h	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$
2^{-0}	5.19e-3	3.65e-09	2.18e-2	3.88e-09	7.18e-2	2.98e-09	2.18e-2	1.31e-08	2.18e-2	4.57e-09
2^{-1}	9.75e-4	8.16e-09	7.74e-3	9.00e-09	4.68e-2	6.80e-09	8.53e-3	1.33e-08	7.76e-3	1.05e-08
2^{-2}	1.62e-4	1.40e-08	2.33e-3	1.41e-08	2.77e-2	1.07e-08	2.80e-3	1.64e-08	2.33e-3	1.50e-08
2^{-3}	2.35e-5	5.33e-09	6.41e-4	5.40e-09	1.53e-2	5.71e-09	8.28e-4	6.13e-09	6.38e-4	7.05e-09
2^{-4}	1.77e-5	6.53e-09	1.54e-4	6.57e-09	8.01e-3	6.95e-09	2.11e-4	7.79e-09	1.49e-4	7.53e-09
2^{-5}	6.11e-6	3.72e-09	3.67e-5	3.73e-09	4.12e-3	4.79e-09	5.77e-5	5.29e-09	3.16e-5	4.50e-09

TABLE 6.4
Results for the approximation of $\mathrm{E}(X_t^1 X_t^2)$ *for SDE* (6.2).

	RI5		RI6		EM		ExEu		PL1WM	
h	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$	$\lvert\hat{\mu}\rvert$	$\hat{\sigma}_\mu^2$
2^{-0}	2.61e-2	5.21e-09	5.03e-2	4.34e-09	3.27e-1	8.15e-10	2.43e-1	6.46e-09	3.92e-2	2.83e-09
2^{-1}	2.20e-2	1.45e-08	2.98e-2	6.24e-09	2.85e-1	4.33e-09	4.85e-2	1.45e-08	2.73e-2	5.34e-09
2^{-2}	8.32e-3	3.55e-08	1.11e-2	1.76e-08	1.67e-1	6.23e-09	6.60e-3	2.84e-08	1.12e-2	9.88e-09
2^{-3}	2.17e-3	7.24e-08	3.03e-3	7.73e-08	8.67e-2	2.19e-08	1.75e-3	1.05e-07	3.36e-3	8.05e-08
2^{-4}	5.54e-4	1.58e-07	7.09e-4	1.37e-07	4.42e-2	3.46e-08	4.63e-4	1.85e-07	8.18e-4	1.86e-07
2^{-5}	3.03e-4	3.14e-07	3.81e-4	3.16e-07	2.24e-2	1.31e-07	3.07e-4	3.08e-07	4.05e-4	3.07e-07

mutative noise given by

(6.2)
$$\mathrm{d}\begin{pmatrix} X_t^1 \\ X_t^2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}X_t^1 + \frac{3}{2}X_t^2 \\ \frac{3}{2}X_t^1 - \frac{1}{2}X_t^2 \end{pmatrix} \mathrm{d}t + \begin{pmatrix} \sqrt{\frac{3}{4}(X_t^1)^2 - \frac{3}{2}X_t^1 X_t^2 + \frac{3}{4}(X_t^2)^2 + \frac{3}{20}} \\ 0 \end{pmatrix} \mathrm{d}W_t^1$$
$$+ \begin{pmatrix} -\sqrt{\frac{1}{4}(X_t^1)^2 - \frac{1}{2}X_t^1 X_t^2 + \frac{1}{4}(X_t^2)^2 + \frac{1}{20}} \\ \sqrt{(X_t^1)^2 - 2X_t^1 X_t^2 + X_t^2 + \frac{1}{5}} \end{pmatrix} \mathrm{d}W_t^2,$$

with initial value $X_0 = (\frac{1}{10}, \frac{1}{10})^T$. Then the corresponding moments of the solution are $\mathrm{E}(X_t^i) = \frac{1}{10}\exp(t)$, $\mathrm{E}((X_t^i)^2) = \frac{3}{50}\exp(2t) - \frac{1}{10}\exp(-t) + \frac{1}{20}$, and $\mathrm{E}(X_t^1 X_t^2) = \frac{3}{50}\exp(2t) + \frac{1}{5}\exp(-t) - \frac{1}{4}$ for $i = 1, 2$. Here, we choose $M_1 = 20$ and $M_2 = 5 \times 10^7$. The corresponding results are presented in Figures 6.2–6.3 and Tables 6.3–6.4.

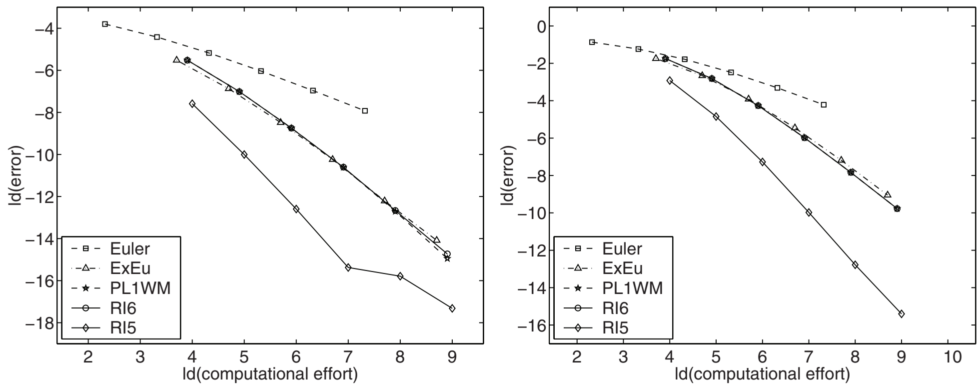

FIG. 6.3. *Computational effort vs. error for the approximation of* $\mathrm{E}(X_T^1)$ *for SDE* (6.2) *(left) and for SDE* (6.3) *for* $\lambda = \mu = 0$ *with* $m = 2$ *(right).*

Finally, we consider a nonlinear SDE with noncommutative noise and some higher dimension $d = 4$ which is given for $\lambda, \mu \in \{0, 1\}$ as

(6.3)

$$\mathrm{d}\begin{pmatrix} X_t^1 \\ X_t^2 \\ X_t^3 \\ X_t^4 \end{pmatrix} = \begin{pmatrix} \frac{243}{154}X_t^1 - \frac{27}{77}X_t^2 + \frac{23}{154}X_t^3 - \frac{65}{154}X_t^4 \\ \frac{27}{77}X_t^1 - \frac{243}{154}X_t^2 + \frac{65}{154}X_t^3 - \frac{23}{154}X_t^4 \\ \frac{5}{154}X_t^1 - \frac{61}{154}X_t^2 + \frac{162}{77}X_t^3 - \frac{36}{77}X_t^4 \\ \frac{61}{154}X_t^1 - \frac{5}{154}X_t^2 + \frac{36}{77}X_t^3 - \frac{162}{77}X_t^4 \end{pmatrix} \mathrm{d}t$$

$$+ \frac{1}{9}\sqrt{(X_t^2)^2 + (X_t^3)^2 + \frac{2}{23}} \begin{pmatrix} \frac{1}{13} \\ \frac{1}{14} \\ \frac{1}{13} \\ \frac{1}{15} \end{pmatrix} \mathrm{d}W_t^1 + \frac{1}{8}\sqrt{(X_t^4)^2 + (X_t^1)^2 + \frac{1}{11}} \begin{pmatrix} \frac{1}{14} \\ \frac{1}{16} \\ \frac{1}{16} \\ \frac{1}{12} \end{pmatrix} \mathrm{d}W_t^2$$

$$+ \frac{\lambda}{12}\sqrt{(X_t^1)^2 + (X_t^2)^2 + \frac{1}{9}} \begin{pmatrix} \frac{1}{6} \\ \frac{1}{5} \\ \frac{1}{5} \\ \frac{1}{6} \end{pmatrix} \mathrm{d}W_t^3 + \frac{\lambda}{14}\sqrt{(X_t^3)^2 + (X_t^4)^2 + \frac{3}{29}} \begin{pmatrix} \frac{1}{8} \\ \frac{1}{9} \\ \frac{1}{8} \\ \frac{1}{9} \end{pmatrix} \mathrm{d}W_t^4$$

$$+ \frac{\mu}{10}\sqrt{(X_t^1)^2 + (X_t^3)^2 + \frac{1}{13}} \begin{pmatrix} \frac{1}{11} \\ \frac{1}{15} \\ \frac{1}{13} \\ \frac{1}{11} \end{pmatrix} \mathrm{d}W_t^5 + \frac{\mu}{11}\sqrt{(X_t^2)^2 + (X_t^4)^2 + \frac{2}{25}} \begin{pmatrix} \frac{1}{12} \\ \frac{1}{13} \\ \frac{1}{16} \\ \frac{1}{13} \end{pmatrix} \mathrm{d}W_t^6$$

with initial value $X_0 = (\frac{1}{8}, \frac{1}{8}, 1, \frac{1}{8})^T$. Then we have $m = 2 + 2\lambda + 2\mu$ as the dimension of the driving Wiener process. The moments of the solution can be calculated as $\mathrm{E}(X_T^i) = \frac{1}{8}\exp(2T)$ for $i = 1, 2, 4$ and $\mathrm{E}(X_T^3) = \exp(2T)$. Here, we compare the performance of the considered schemes for the cases $m = 2$ with $\lambda = \mu = 0$, for $m = 4$ with $\lambda = 1$ and $\mu = 0$, and for $m = 6$ if $\lambda = \mu = 1$. The results are presented in Figures 6.3–6.4 for $m = 2, 4, 6$ and in Table 6.5 for the case of $\lambda = \mu = 1$.

On the right-hand sides in Figure 6.3 and in Figure 6.4, we can see the performance of the considered schemes as the dimension m increases from 2 to 6. Comparing these

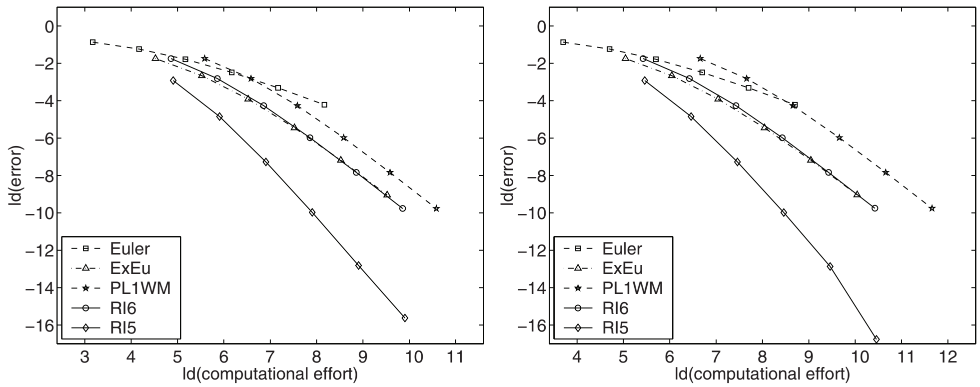

FIG. 6.4. *Computational effort vs. error for the approximation of* $\mathrm{E}(X_T^1)$ *for SDE* (6.3) *for* $\lambda = 1$, $\mu = 0$ *with* $m = 4$ *(left) and for* $\lambda = \mu = 1$ *with* $m = 6$ *(right).*

TABLE 6.5
Results for the approximation of $\mathrm{E}(X_t^1)$ *for SDE* (6.3) *with* $\lambda = \mu = 1$.

	RI5		RI6		EM		ExEu		PL1WM	
h	$\|\hat{\mu}\|$	$\hat{\sigma}_\mu^2$	$\|\hat{\mu}\|$	$\hat{\sigma}_\mu^2$	$\|\hat{\mu}\|$	$\hat{\sigma}_\mu^2$	$\|\hat{\mu}\|$	$\hat{\sigma}_\mu^2$	$\|\hat{\mu}\|$	$\hat{\sigma}_\mu^2$
2^{-0}	1.32e-1	9.30e-10	2.99e-1	1.07e-09	5.49e-1	1.93e-10	2.99e-1	1.07e-09	2.99e-1	1.07e-09
2^{-1}	3.47e-2	3.01e-09	1.42e-1	2.85e-09	4.24e-1	6.19e-10	1.58e-1	2.60e-09	1.42e-1	2.87e-09
2^{-2}	6.46e-3	4.63e-09	5.20e-2	4.41e-09	2.91e-1	1.51e-09	6.63e-2	4.14e-09	5.20e-2	4.48e-09
2^{-3}	9.87e-4	4.80e-09	1.58e-2	4.75e-09	1.79e-1	2.50e-09	2.29e-2	4.73e-09	1.58e-2	4.86e-09
2^{-4}	1.24e-4	4.76e-09	4.38e-3	4.78e-09	1.01e-1	3.40e-09	6.88e-3	4.88e-09	4.38e-3	4.93e-09
2^{-5}	1.21e-5	5.63e-09	1.15e-3	5.64e-09	5.38e-2	4.78e-09	1.90e-3	5.77e-09	1.15e-3	5.80e-09

results, we can see the significantly reduced complexity for the new SRK schemes RI5 and RI6 compared to the well-known SRK scheme PL1WM in the case of $m > 2$. This benefit becomes more and more significant if we increase the dimension m of the driving Wiener process, which confirms our theoretical results. For the considered examples, we obtained very good results especially for the SRK scheme RI5 having order $p_D = 3$ and $p_S = 2$.

7. Conclusion. The main advantage of the introduced class of SRK methods (5.4) is the significant reduction of the computational costs. Considering, for example, the order two SRK schemes proposed in [7, p. 486] and in [24] for Itô SDEs with a multidimensional Wiener process, at least 2 evaluations of the drift function a and $2m+1$ evaluations of each diffusion function b^k, $k = 1, \ldots, m$, are necessary for each step. Further, $m(m+1)/2$ independent random variables have to be simulated for the schemes in [7, 24] for each step. Nearly the same applies for the class of SRK schemes recently proposed in [9] for Stratonovich SDEs, which need 4 evaluations of the drift a and $3m+1$ evaluations of each diffusion function b^k, $k = 1, \ldots, m$, in each step. Due to the dependence of the computational costs on the dimension m of the driving Wiener process, these SRK methods are not of much relevance in practice, especially for high-dimensional problems. In contrast to this, the new SRK scheme RI6 needs 2 evaluations of the drift function a and only 5 evaluations of the diffusion function b^k, $k = 1, \ldots, m$, for each step due to $s = 3$ stages and $H_1^{(k)} = \hat{H}_1^{(k)}$. Further, only $2m - 1$ independent random variables have to be simulated for each step. The computational effort of the new SRK schemes is even similar to the one of the extrapolated Euler–Maruyama scheme, where 3 evaluations of the drift a and 3 evaluations of the diffusion functions b^k, $k = 1, \dots, m$, are necessary, and which needs the simulation of $2m$ independent random variables at each step. However, there are situations where extrapolation methods, especially based on the Euler–Maruyama scheme, have only limited value [12]. This is the case, e.g., for stiff problems due to the restricted stability regions of the Euler–Maruyama method [8]. Thus, as in the deterministic setting, higher order one-step methods such as the introduced SRK methods are of independent relevance.

As a result of this, the introduced class of SRK methods is of considerable importance, now also for high-dimensional problems such as those in mathematical finance or physics. Future research may be done by developing efficient SRK schemes of some higher order than order two. Further, a stability analysis and the investigation of implicit SRK methods is of particular interest. Similar to the deterministic setting, embedded SRK schemes can be easily implemented which can be applied with a step size control algorithm; see also [11, 13].

Appendix. Proof of Theorem 5.1. First, we have to prove that the SRK method (5.4) is contained in the class of SRK methods (2.1). Therefore, choose $\mathcal{M} = \{(k), (k,l) : 0 \leq k, l \leq m\}$ and

$$\gamma_i^{(\iota)(k,\nu)} \theta_\iota(h_n) = \begin{cases} \beta_i^{(1)} \hat{I}_{(k)} & \text{if } 0 < \iota = k, \nu = 0, \\ \beta_i^{(2)} \frac{\hat{I}_{(k,k)}}{\sqrt{h_n}} & \text{if } \iota = (k,k), 0 < k, \nu = 0, \\ \beta_i^{(3)} \hat{I}_{(k)} & \text{if } 0 < \iota = k, \nu = 1, \\ \beta_i^{(4)} \sqrt{h_n} & \text{if } 0 = \iota < k, \nu = 1, \\ 0 & \text{otherwise}, \end{cases}$$

$$A_{ij}^{(k,\nu)(0,0)} h_n = \begin{cases} A_{ij}^{(0)} h_n & \text{if } k = \nu = 0, \\ A_{ij}^{(1)} h_n & \text{if } k > 0, \nu = 0, \\ A_{ij}^{(2)} h_n & \text{if } k > 0, \nu = 1, \\ 0 & \text{otherwise}, \end{cases}$$

$$B_{ij}^{(\iota)(k,\nu)(r,\mu)} \theta_\iota(h_n) = \begin{cases} B_{ij}^{(0)} \hat{I}_{(r)} & \text{if } k = \nu = \mu = 0, \iota = r, \\ B_{ij}^{(1)} \sqrt{h_n} & \text{if } 0 = \iota < k = r, \nu = \mu = 0, \\ B_{ij}^{(2)} \frac{\hat{I}_{(k,r)}}{\sqrt{h_n}} & \text{if } \iota = (k,r), 0 < k \neq r, \nu = 1, \mu = 0, \\ 0 & \text{otherwise} \end{cases}$$

for $k = 0, 1, \dots, m$, $r = 1, \dots, m$, and $\iota, \nu, \mu \in \mathcal{M}$. As a result of this we get

$$z_i^{(0,0)} = \alpha_i \, h_n, \qquad z_i^{(k,0)} = \beta_i^{(1)} \, \hat{I}_{(k)} + \beta_i^{(2)} \, \frac{\hat{I}_{(k,k)}}{\sqrt{h_n}}, \quad z_i^{(k,1)} = \beta_i^{(3)} \, \hat{I}_{(k)} + \beta_i^{(4)} \, \sqrt{h_n},$$
$$Z_{ij}^{(0,0)(0,0)} = A_{ij}^{(0)} \, h_n, \quad Z_{ij}^{(k,0)(0,0)} = A_{ij}^{(1)} \, h_n, \qquad Z_{ij}^{(k,1)(0,0)} = A_{ij}^{(2)} \, h_n,$$
$$Z_{ij}^{(0,0)(k,0)} = B_{ij}^{(0)} \hat{I}_{(k)}, \quad Z_{ij}^{(k,0)(l,0)} = B_{ij}^{(1)} \sqrt{h_n} \mathbf{1}_{\{k=l\}}, \quad Z_{ij}^{(k,1)(l,0)} = B_{ij}^{(2)} \frac{\hat{I}_{(k,l)}}{\sqrt{h_n}} \mathbf{1}_{\{k \neq l\}}$$

for $1 \leq k, l \leq m$. Furthermore, we have $H_i^{(0,0)} = H_i^{(0)}$, $H_i^{(k,0)} = H_i^{(k)}$, and $H_i^{(k,1)} = \hat{H}_i^{(k)}$. Thus, the SRK method (5.4) is covered by the class (2.1).

Next, in order to prove Theorem 5.1 we have to calculate the order two conditions for the SRK method (5.4). Therefore, we apply Theorem 4.2 with $p = 2$; i.e., we have to consider all trees $\mathbf{t} \in TS(\Delta)$ with $\rho(\mathbf{t}) \leq 2.5$. We refer to [18] for all necessary trees and corresponding parameters $\alpha_I(\mathbf{t})$, $\alpha_\Delta(\mathbf{t})$, or $\beta(\mathbf{t})$. Thus, we have to calculate the conditions (4.1) for each tree. Apart from (5.6) and (5.7), the following moments are helpful in the subsequent calculations: $\mathrm{E}(\hat{I}^2_{(k,k)}) = \frac{1}{2}h^2$, $\mathrm{E}(\hat{I}^2_{(k)} \hat{I}_{(k,k)}) = h^2$, $\mathrm{E}(\hat{I}^3_{(k,k)}) = h^3$, $\mathrm{E}(\hat{I}_{(k)} \hat{I}_{(l)} \hat{I}_{(k,l)}) = \frac{1}{2}h^2$, $\mathrm{E}(\hat{I}^2_{(k,l)}) = \frac{1}{2}h^2$, $\mathrm{E}(\hat{I}^q_{(k,l)}) = 0$, $\mathrm{E}(\hat{I}^2_{(k)} \hat{I}^2_{(k,l)}) = h^3$, and $\mathrm{E}(\hat{I}_{(k,l)} \hat{I}_{(l,k)}) = 0$ for $q = 1, 3$ and $k \neq l$. In the following, $\beta(\mathbf{t}) = 1$ is fulfilled if not stated otherwise.

Order 0.5 *trees.* $\mathbf{t}_{0.5,1} = (\sigma_{j_1})$: $\Phi_S(\mathbf{t}_{0.5,1}) = z^{(j_1,0)^T} e + z^{(j_1,1)^T} e$. With $\alpha_I(\mathbf{t}_{0.5,1}) = 0$ we calculate from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{0.5,1})) = 0$, which is fulfilled if $\beta^{(4)^T} e \sqrt{h} = 0$.

Therefore, we assume that condition 2 of Theorem 5.1 is fulfilled in the following.

Order 1.0 *trees.* $\mathbf{t}_{1,1} = (\tau)$: $\Phi_S(\mathbf{t}_{1,1}) = z^{(0,0)^T} e$. With $\alpha_I(\mathbf{t}_{1,1}) = \alpha_\Delta(\mathbf{t}_{1,1}) = 1$ we obtain from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{1,1})) = h$, which is fulfilled if $\alpha^T e\, h = h$.

$\mathbf{t}_{1,2} = (\sigma_{j_1}, \sigma_{j_2})$: $\Phi_S(\mathbf{t}_{1,2}) = (z^{(j_1,0)^T} e + z^{(j_1,1)^T} e)(z^{(j_2,0)^T} e + z^{(j_2,1)^T} e)$. For $j_1 = j_2$ with $\alpha_I(\mathbf{t}_{1,2}) = \alpha_\Delta(\mathbf{t}_{1,2}) = 1$ we get from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{1,2})) = h$, which is fulfilled if $(\beta^{(1)^T} e)^2\ \mathrm{E}(\hat{I}^2_{(j_1)}) + (\beta^{(2)^T} e)^2\ \mathrm{E}(\hat{I}^2_{(j_1,j_1)}) h^{-1} + (\beta^{(3)^T} e)^2\ \mathrm{E}(\hat{I}^2_{(j_1)}) + (\beta^{(1)^T} e)\, (\beta^{(3)^T} e)\ \mathrm{E}(\hat{I}^2_{(j_1)}) = h$.

$\mathbf{t}_{1,3} = (\{\sigma_{j_2}\}_{j_1})$: $\Phi_S(\mathbf{t}_{1,3}) = z^{(j_1,0)^T} Z^{(j_1,0)(j_2,0)} e + z^{(j_1,1)^T} Z^{(j_1,1)(j_2,0)} e$. Here, we have $\alpha_I(\mathbf{t}_{1,3}) = 0$, and (4.1) is fulfilled due to $\mathrm{E}(\Phi_S(\mathbf{t}_{1,3})) = 0$.

Now, we additionally assume that condition 1 of Theorem 5.1 is fulfilled.

Order 1.5 *trees.* $\mathbf{t}_{1.5,2} = (\{\tau\}_{j_1})$: $\Phi_S(\mathbf{t}_{1.5,2}) = z^{(j_1,0)^T} Z^{(j_1,0)(0,0)} e + z^{(j_1,1)^T} Z^{(j_1,1)(0,0)} e$. We calculate with $\alpha_I(\mathbf{t}_{1.5,2}) = 0$ from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{1.5,2})) = 0$ if $\beta^{(4)^T} A^{(2)} e\, h^{3/2} = 0$ holds.

$\mathbf{t}_{1.5,4} = (\sigma_{j_1}, \sigma_{j_2}, \sigma_{j_3})$: $\Phi_S(\mathbf{t}_{1.5,4}) = (z^{(j_1,0)^T} e + z^{(j_1,1)^T} e)\, (z^{(j_2,0)^T} e + z^{(j_2,1)^T} e) \times (z^{(j_3,0)^T} e + z^{(j_3,1)^T} e)$. For $j_1 = j_2 = j_3$ with $\alpha_I(\mathbf{t}_{1.5,4}) = 0$ we calculate from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{1.5,4})) = 0$ is fulfilled if $(\beta^{(1)^T} e + \beta^{(3)^T} e)^2\, (\beta^{(2)^T} e)\ \mathrm{E}(\hat{I}^2_{(j_1)} \hat{I}_{(j_1,j_1)}) h^{-1/2} + (\beta^{(2)^T} e)^3\ \mathrm{E}(\hat{I}^3_{(j_1,j_1)}) h^{-3/2} = 0$. Thus, we need $\beta^{(2)^T} e = 0$.

In the following, due to $\mathbf{t}_{1,2}$ we assume $(\beta^{(1)^T} e)^2 = 1$ and $\beta^{(3)^T} e = 0$.

$\mathbf{t}_{1.5,5} = (\{\sigma_{j_2}\}_{j_1}, \sigma_{j_3})$: $\Phi_S(\mathbf{t}_{1.5,5}) = (z^{(j_1,0)^T} Z^{(j_1,0)(j_2,0)} e + z^{(j_1,1)^T} Z^{(j_1,1)(j_2,0)} e) \times (z^{(j_3,0)^T} e + z^{(j_3,1)^T} e)$. Case (A): For $j_1 = j_2 = j_3$ with $\alpha_I(\mathbf{t}_{1.5,5}) = 0$ condition (4.1) yields $\mathrm{E}(\Phi_S(\mathbf{t}_{1.5,5})) = 0$, which is fulfilled if $(\beta^{(1)^T} e)\, (\beta^{(1)^T} B^{(1)} e)\ \mathrm{E}(\hat{I}^2_{(j_1)}) \sqrt{h} = 0$. Case (B): For $j_1 \neq j_2 = j_3$ with $\alpha_I(\mathbf{t}_{1.5,5}) = 0$ we calculate from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{1.5,5})) = 0$, which is fulfilled if $(\beta^{(1)^T} e)\, (\beta^{(3)^T} B^{(2)} e)\ \mathrm{E}(\hat{I}_{(j_1)} \hat{I}_{(j_2)} \hat{I}_{(j_1,j_2)}) h^{-1/2} = 0$.

$\mathbf{t}_{1.5,6} = (\{\sigma_{j_2}, \sigma_{j_3}\}_{j_1})$: $\Phi_S(\mathbf{t}_{1.5,6}) = z^{(j_1,0)^T} ((Z^{(j_1,0)(j_2,0)} e)(Z^{(j_1,0)(j_3,0)} e)) + z^{(j_1,1)^T} \times ((Z^{(j_1,1)(j_2,0)} e)(Z^{(j_1,1)(j_3,0)} e))$. For $j_1 \neq j_2 = j_3$ with $\alpha_I(\mathbf{t}_{1.5,6}) = 0$ we get from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{1.5,6})) = 0$, which is fulfilled if $\beta^{(4)^T} (B^{(2)} e)^2\ \mathrm{E}(\hat{I}^2_{(j_1,j_2)}) h^{-1/2} = 0$.

For the trees $\mathbf{t}_{1.5,1} = ([\sigma_{j_1}])$, $\mathbf{t}_{1.5,3} = (\tau, \sigma_{j_1})$, and $\mathbf{t}_{1.5,7} = (\{\{\sigma_{j_3}\}_{j_2}\}_{j_1})$ we have $\alpha_I(\mathbf{t}) = 0$ and $\mathrm{E}(\Phi_S(\mathbf{t})) = 0$ for any choice of the coefficients.

Order 2.0 *trees.* $\mathbf{t}_{2,1} = ([\tau])$: $\Phi_S(\mathbf{t}_{2,1}) = z^{(0,0)^T} Z^{(0,0)(0,0)} e$. With $\alpha_I(\mathbf{t}_{2,1}) = \alpha_\Delta(\mathbf{t}_{2,1}) = 1$ we obtain from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,1})) = \frac{1}{2}h^2$, which is fulfilled if $\alpha^T A^{(0)} e\, h^2 = \frac{1}{2}h^2$.

$\mathbf{t}_{2,2} = (\tau,\tau)$: $\Phi_S(\mathbf{t}_{2,2}) = ({z^{(0,0)}}^T e)^2$. With $\alpha_I(\mathbf{t}_{2,2}) = \alpha_\Delta(\mathbf{t}_{2,2}) = 1$ we calculate from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,2})) = h^2$, which is fulfilled if $(\alpha^T e)^2\, h^2 = h^2$.

$\mathbf{t}_{2,4} = ([\sigma_{j_1},\sigma_{j_2}])$: $\Phi_S(\mathbf{t}_{2,4}) = {z^{(0,0)}}^T((Z^{(0,0)(j_1,0)}e)(Z^{(0,0)(j_2,0)}e))$. For $j_1 = j_2$ with $\alpha_I(\mathbf{t}_{2,4}) = \alpha_\Delta(\mathbf{t}_{2,4}) = 1$ we get from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,4})) = \frac{1}{2}h^2$, which is fulfilled if $\alpha^T(B^{(0)}e)^2\ \mathrm{E}(\hat{I}^2_{(j_1)})h = \frac{1}{2}h^2$.

$\mathbf{t}_{2,5} = (\sigma_{j_1},[\sigma_{j_2}])$: $\Phi_S(\mathbf{t}_{2,5}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)\,({z^{(0,0)}}^T Z^{(0,0)(j_2,0)}e)$. For $j_1 = j_2$ with $\alpha_I(\mathbf{t}_{2,5}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,5}) = 3$ we calculate from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,5})) = \frac{1}{2}h^2$, which is fulfilled if $({\beta^{(1)}}^T e)\,(\alpha^T B^{(0)}e)\ \mathrm{E}(\hat{I}^2_{(j_1)})h = \frac{1}{2}h^2$.

$\mathbf{t}_{2,7} = (\sigma_{j_1},\sigma_{j_2},\tau)$: $\Phi_S(\mathbf{t}_{2,7}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)({z^{(j_2,0)}}^T e + {z^{(j_2,1)}}^T e)({z^{(0,0)}}^T e)$. For $j_1 = j_2$ with $\alpha_I(\mathbf{t}_{2,7}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,7}) = 3$ we calculate from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,7})) = h^2$, which is fulfilled if $({\beta^{(1)}}^T e)^2(\alpha^T e)\ \mathrm{E}(\hat{I}^2_{(j_1)})h = h^2$.

$\mathbf{t}_{2,8} = (\sigma_{j_1},\{\tau\}_{j_2})$: $\Phi_S(\mathbf{t}_{2,8}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)({z^{(j_2,0)}}^T Z^{(j_2,0)(0,0)}e + {z^{(j_2,1)}}^T \times Z^{(j_2,1)(0,0)}e)$. For $j_1 = j_2$ with $\alpha_I(\mathbf{t}_{2,8}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,8}) = 3$ we obtain from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,8})) = \frac{1}{2}h^2$, which is fulfilled if $({\beta^{(1)}}^T e)({\beta^{(1)}}^T A^{(1)}e + {\beta^{(3)}}^T A^{(2)}e)\ \mathrm{E}(\hat{I}^2_{(j_1)})h = \frac{1}{2}h^2$.

Here, we claim that $({\beta^{(1)}}^T e)({\beta^{(1)}}^T A^{(1)}e) = \frac{1}{2}$ and ${\beta^{(3)}}^T A^{(2)}e = 0$.

$\mathbf{t}_{2,11} = (\sigma_{j_1},\sigma_{j_2},\sigma_{j_3},\sigma_{j_4})$: $\Phi_S(\mathbf{t}_{2,11}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)({z^{(j_2,0)}}^T e + {z^{(j_2,1)}}^T e) \times ({z^{(j_3,0)}}^T e + {z^{(j_3,1)}}^T e)({z^{(j_4,0)}}^T e + {z^{(j_4,1)}}^T e)$. Case (A): For $j_1 = j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,11}) = \alpha_\Delta(\mathbf{t}_{2,11}) = \beta(\mathbf{t}_{2,11}) = 1$ we get from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,11})) = 3h^2$, which is fulfilled if $({\beta^{(1)}}^T e)^4\ \mathrm{E}(\hat{I}^4_{(j_1)}) = 3h^2$. Case (B): For $j_1 = j_2 \neq j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,11}) = \alpha_\Delta(\mathbf{t}_{2,11}) = 1$ and $\beta(\mathbf{t}_{2,11}) = 3$ we calculate from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,11})) = h^2$, which is fulfilled if $({\beta^{(1)}}^T e)^2({\beta^{(1)}}^T e)^2\ \mathrm{E}(\hat{I}^2_{(j_1)}\hat{I}^2_{(j_3)}) = h^2$.

$\mathbf{t}_{2,12} = (\sigma_{j_1},\sigma_{j_2},\{\sigma_{j_4}\}_{j_3})$: $\Phi_S(\mathbf{t}_{2,12}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)({z^{(j_2,0)}}^T e + {z^{(j_2,1)}}^T e) \times ({z^{(j_3,0)}}^T Z^{(j_3,0)(j_4,0)}e + {z^{(j_3,1)}}^T Z^{(j_3,1)(j_4,0)}e)$. Case (A): For $j_1 = j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,12}) = 4$, $\alpha_\Delta(\mathbf{t}_{2,12}) = 6$ and $\beta(\mathbf{t}_{2,12}) = 1$ we obtain from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) = h^2$, which is fulfilled if $({\beta^{(1)}}^T e)^2({\beta^{(2)}}^T B^{(1)}e)\ \mathrm{E}(\hat{I}^2_{(j_1)}\hat{I}_{(j_1,j_1)}) = h^2$. Case (B): For $j_1 = j_3 \neq j_2 = j_4$ with $\alpha_I(\mathbf{t}_{2,12}) = 4$, $\alpha_\Delta(\mathbf{t}_{2,12}) = 6$, and $\beta(\mathbf{t}_{2,12}) = 2$ we calculate from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,12})) = \frac{1}{2}h^2$, which is fulfilled if $({\beta^{(1)}}^T e)^2({\beta^{(4)}}^T B^{(2)}e)\ \mathrm{E}(\hat{I}_{(j_1)}\hat{I}_{(j_2)}\hat{I}_{(j_1,j_2)}) = \frac{1}{2}h^2$.

$\mathbf{t}_{2,13} = (\sigma_{j_1},\{\sigma_{j_3},\sigma_{j_4}\}_{j_2})$: $\Phi_S(\mathbf{t}_{2,13}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)({z^{(j_2,0)}}^T((Z^{(j_2,0)(j_3,0)}e) \times (Z^{(j_2,0)(j_4,0)}e)) + {z^{(j_2,1)}}^T((Z^{(j_2,1)(j_3,0)}e)(Z^{(j_2,1)(j_4,0)}e)))$. Case (A): For $j_1 = j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,13}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,13}) = 4$ formula (4.1) yields $\mathrm{E}(\Phi_S(\mathbf{t}_{2,13})) = \frac{1}{2}h^2$, which is fulfilled if $({\beta^{(1)}}^T e)({\beta^{(1)}}^T(B^{(1)}e)^2)\ \mathrm{E}(\hat{I}^2_{(j_1)})h = \frac{1}{2}h^2$. Case (B): For $j_1 = j_2 \neq j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,13}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,13}) = 4$ we obtain from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{2,13})) = \frac{1}{2}h^2$, which is fulfilled if $({\beta^{(1)}}^T e)({\beta^{(3)}}^T(B^{(2)}e)^2)\ \mathrm{E}(\hat{I}^2_{(j_1)}\hat{I}^2_{(j_1,j_3)})h^{-1} = \frac{1}{2}h^2$.

$\mathbf{t}_{2,14} = (\sigma_{j_1},\{\{\sigma_{j_4}\}_{j_3}\}_{j_2})$: $\Phi_S(\mathbf{t}_{2,14}) = ({z^{(j_1,0)}}^T e + {z^{(j_1,1)}}^T e)({z^{(j_2,0)}}^T(Z^{(j_2,0)(j_3,0)} \times (Z^{(j_3,0)(j_4,0)}e)) + {z^{(j_2,1)}}^T(Z^{(j_2,1)(j_3,0)}(Z^{(j_3,0)(j_4,0)}e)))$. Case (A): For $j_1 = j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,14}) = 0$ we calculate from (4.1) the condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,14})) = 0$, which is fulfilled if $({\beta^{(1)}}^T e)({\beta^{(1)}}^T(B^{(1)}(B^{(1)}e)))\ \mathrm{E}(\hat{I}^2_{(j_1)})h = 0$. Case (B): For $j_1 = j_3 = j_4 \neq j_2$ with $\alpha_I(\mathbf{t}_{2,14}) = 0$ we obtain from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,14})) = 0$, which is fulfilled if $({\beta^{(1)}}^T e)({\beta^{(3)}}^T(B^{(2)}(B^{(1)}e)))\ \mathrm{E}(\hat{I}_{(j_1)}\hat{I}_{(j_2)}\hat{I}_{(j_2,j_1)}) = 0$.

$\mathbf{t}_{2,15} = (\{\sigma_{j_2}\}_{j_1}, \{\sigma_{j_4}\}_{j_3})$: $\Phi_S(\mathbf{t}_{2,15}) = (z^{(j_1,0)^T} Z^{(j_1,0)(j_2,0)} e + z^{(j_1,1)^T} Z^{(j_1,1)(j_2,0)} e) \times (z^{(j_3,0)^T} Z^{(j_3,0)(j_4,0)} e + z^{(j_3,1)^T} Z^{(j_3,1)(j_4,0)} e)$. Case (A): For $j_1 = j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,15}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,15}) = 3$ we get from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) = \frac{1}{2}h^2$ is fulfilled if $(\beta^{(1)^T} B^{(1)} e)^2 \, \mathrm{E}(\hat{I}^2_{(j_1)})h + (\beta^{(2)^T} B^{(1)} e)^2 \, \mathrm{E}(\hat{I}^2_{(j_1,j_1)}) = \frac{1}{2}h^2$. Case (B): For $j_1 = j_3 \neq j_2 = j_4$ with $\alpha_I(\mathbf{t}_{2,15}) = 2$ and $\alpha_\Delta(\mathbf{t}_{2,15}) = 3$ we obtain from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,15})) = \frac{1}{2}h^2$, which is fulfilled if $(\beta^{(4)^T} B^{(2)} e)^2 \, \mathrm{E}(\hat{I}^2_{(j_1,j_2)}) + (\beta^{(3)^T} B^{(2)} e)^2 \, \mathrm{E}(\hat{I}^2_{(j_1)} \hat{I}^2_{(j_1,j_2)}) h^{-1} = \frac{1}{2}h^2$.

$\mathbf{t}_{2,17} = (\{\sigma_{j_2}, \{\sigma_{j_4}\}_{j_3}\}_{j_1})$: $\Phi_S(\mathbf{t}_{2,17}) = z^{(j_1,0)^T}((Z^{(j_1,0)(j_2,0)} e)(Z^{(j_1,0)(j_3,0)}(Z^{(j_3,0)(j_4,0)} \times e))) + z^{(j_1,1)^T}((Z^{(j_1,1)(j_2,0)} e)(Z^{(j_1,1)(j_3,0)}(Z^{(j_3,0)(j_4,0)} e)))$. For $j_1 \neq j_2 = j_3 = j_4$ with $\alpha_I(\mathbf{t}_{2,17}) = 0$ we calculate from (4.1) the order condition $\mathrm{E}(\Phi_S(\mathbf{t}_{2,17})) = 0$, which is fulfilled if $\beta^{(4)^T}((B^{(2)} e)(B^{(2)}(B^{(1)} e))) \, \mathrm{E}(\hat{I}^2_{(j_1,j_2)}) = 0$.

$\mathbf{t}_{2,20} = (\{[\sigma_{j_2}]\}_{j_1})$: $\Phi_S(\mathbf{t}_{2,20}) = z^{(j_1,0)^T}(Z^{(j_1,0)(0,0)}(Z^{(0,0)(j_2,0)} e)) + z^{(j_1,1)^T} (Z^{(j_1,1)(0,0)}(Z^{(0,0)(j_2,0)} e))$. For $j_1 = j_2$ with $\alpha_I(\mathbf{t}_{2,20}) = 0$ we get from (4.1) that $\mathrm{E}(\Phi_S(\mathbf{t}_{2,20})) = 0$, which is fulfilled if $\beta^{(3)^T}(A^{(2)}(B^{(0)} e)) \, \mathrm{E}(\hat{I}^2_{(j_1)}) + \beta^{(1)^T}(A^{(1)}(B^{(0)} e)) \, \mathrm{E}(\hat{I}^2_{(j_1)}) = 0$. Here, we claim that $\beta^{(3)^T}(A^{(2)}(B^{(0)} e)) = 0$ and $\beta^{(1)^T}(A^{(1)}(B^{(0)} e)) = 0$.

For all correlations between $j_1, \ldots, j_4$ which have not been considered explicitly, we have $\alpha_I(\mathbf{t}) = 0$ and $\mathrm{E}(\Phi_S(\mathbf{t})) = 0$. Further, for the trees $\mathbf{t}_{2,3} = ([\{\sigma_{j_2}\}_{j_1}])$, $\mathbf{t}_{2,6} = (\{\sigma_{j_2}\}_{j_1}, \tau)$, $\mathbf{t}_{2,9} = (\{\{\tau\}_{j_2}\}_{j_1})$, $\mathbf{t}_{2,10} = (\{\sigma_{j_2}, \tau\}_{j_1})$, $\mathbf{t}_{2,16} = (\{\sigma_{j_2}, \sigma_{j_3}, \sigma_{j_4}\}_{j_1})$, $\mathbf{t}_{2,18} = (\{\{\sigma_{j_3}, \sigma_{j_4}\}_{j_2}\}_{j_1})$, and $\mathbf{t}_{2,19} = (\{\{\{\sigma_{j_4}\}_{j_3}\}_{j_2}\}_{j_1})$ we have $\alpha_I(\mathbf{t}) = 0$, and $\mathrm{E}(\Phi_S(\mathbf{t})) = 0$ is fulfilled without any additional restrictions for the coefficients of the SRK method.

Finally, we have to consider all trees $\mathbf{t} \in TS(\Delta)$ with $\rho(\mathbf{t}) = 2.5$ for which, due to $\alpha_I(\mathbf{t}) = 0$, the order condition $\mathrm{E}(\Phi_S(\mathbf{t})) = 0$ has to be fulfilled. Since the calculations are analogous to those already performed, repetition is avoided (see [18] for all trees up to order 2.5). Leaving out the trees which do not supply any new restrictions for the coefficients, we calculate the following order conditions of Theorem 5.1:

Conditions from $\mathbf{t} \in TS(\Delta)$ with $\rho(\mathbf{t}) = 2.5$.

$\mathbf{t}$	Correlation	Condition
$(\{\tau, \tau\}_{j_1})$		24
$(\{[\tau]\}_{j_1})$		25
$([\{\sigma_{j_2}\}_{j_1}], \sigma_{j_3})$	$j_1 = j_2 = j_3$	26
$([\sigma_{j_1}, \{\sigma_{j_3}\}_{j_2}])$	$j_1 = j_2 = j_3$	47
$(\{\tau\}_{j_1}, \sigma_{j_2}, \sigma_{j_3})$	$j_1 = j_2 = j_3$	27 + 7
	$j_1 \neq j_2 = j_3$	7
$(\{\tau, \sigma_{j_2}\}_{j_1}, \sigma_{j_3})$	$j_1 = j_2 = j_3$	28
	$j_1 \neq j_2 = j_3$	29
$(\{[\sigma_{j_2}]\}_{j_1}, \sigma_{j_3})$	$j_1 = j_2 = j_3$	30 + 31
	$j_1 \neq j_2 = j_3$	30
$(\{\tau, \sigma_{j_2}, \sigma_{j_3}\}_{j_1})$	$j_1 \neq j_2 = j_3$	32
$(\{[\sigma_{j_2}], \sigma_{j_3}\}_{j_1})$	$j_1 = j_2 = j_3$	48
	$j_1 \neq j_2 = j_3$	49
$(\{[\sigma_{j_2}, \sigma_{j_3}]\}_{j_1})$	$j_1 = j_2 = j_3$	33 + 34
	$j_1 \neq j_2 = j_3$	33
$(\{[\{\sigma_{j_3}\}_{j_2}]\}_{j_1})$	$j_1 = j_2 = j_3$	50 + 51
$(\{\{\tau\}_{j_2}\}_{j_1}, \sigma_{j_3})$	$j_1 = j_2 = j_3$	35
	$j_1 \neq j_2 = j_3$	36
$(\{\{\tau\}_{j_2}, \sigma_{j_3}\}_{j_1})$	$j_1 \neq j_2 = j_3$	52
$(\{\{[\sigma_{j_3}]\}_{j_2}\}_{j_1})$	$j_1 = j_2 = j_3$	53
	$j_1 \neq j_2 = j_3$	54

t	Correlation	Condition
$(\sigma_{j_1}, \sigma_{j_2}, \{\sigma_{j_4}, \sigma_{j_5}\}_{j_3})$	$j_1 = j_2 = j_3 = j_4 = j_5$	37
	$j_1 = j_4 \neq j_3,\ j_2 = j_5 \neq j_3$	9
$(\sigma_{j_1}, \sigma_{j_2}, \{\{\sigma_{j_5}\}_{j_4}\}_{j_3})$	$j_1 = j_2 = j_3 = j_4 = j_5$	39
	$j_1 = j_3 \neq j_2 = j_4 = j_5$	38
$(\{\sigma_{j_2}, \sigma_{j_3}, \sigma_{j_4}\}_{j_1}, \sigma_{j_5})$	$j_1 = j_2 = j_3 = j_4 = j_5$	40
	$j_1 \neq j_2 = j_3,\ j_1 \neq j_4 = j_5,\ j_3 \neq j_4$	41
	$j_1 \neq j_2 = j_3 = j_4 = j_5$	41
$(\{\sigma_{j_2}, \{\sigma_{j_4}\}_{j_3}\}_{j_1}, \sigma_{j_5})$	$j_1 = j_2 = j_3 = j_4 = j_5$	55
	$j_1 = j_5 \neq j_2 = j_3 = j_4$	56
$(\{\{\sigma_{j_3}, \sigma_{j_4}\}_{j_2}\}_{j_1}, \sigma_{j_5})$	$j_1 = j_2 = j_3 = j_4 = j_5$	42
	$j_1 \neq j_2 = j_3 = j_4 = j_5$	43
$(\{\{\{\sigma_{j_4}\}_{j_3}\}_{j_2}\}_{j_1}, \sigma_{j_5})$	$j_1 = j_2 = j_3 = j_4 = j_5$	57
	$j_1 \neq j_2 = j_3 = j_4 = j_5$	21
$(\{\sigma_{j_2}, \sigma_{j_3}, \sigma_{j_4}, \sigma_{j_5}\}_{j_1})$	$j_1 \neq j_2 = j_3,\ j_1 \neq j_4 = j_5$	44
$(\{\sigma_{j_2}, \{\sigma_{j_4}, \sigma_{j_5}\}_{j_3}\}_{j_1})$	$j_1 \neq j_2 = j_3 = j_4 = j_5$	59
$(\{\sigma_{j_2}, \{\{\sigma_{j_5}\}_{j_4}\}_{j_3}\}_{j_1})$	$j_1 \neq j_2 = j_3 = j_4 = j_5$	58
$(\{\{\sigma_{j_3}\}_{j_2}, \{\sigma_{j_5}\}_{j_4}\}_{j_1})$	$j_1 \neq j_2 = j_3 = j_4 = j_5$	45

Now, we just have to summarize the calculated conditions in order to obtain Theorem 5.1. Finally, the approximation Y by the SRK method (5.4) has uniformly bounded moments due to $\mathrm{E}(z^{(k,\nu)^T} e) = 0$ for $1 \leq k \leq m$ and $\nu \in \{0, 1\}$. ∎

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